PREFLEX

Product Information Bulletin

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Preflex Ordering Process

Choosing Preflex SIPs means selecting one of the highest-performing building envelopes available today. This choice enhances the energy efficiency, structural strength, and indoor environmental quality of both commercial and residential buildings.

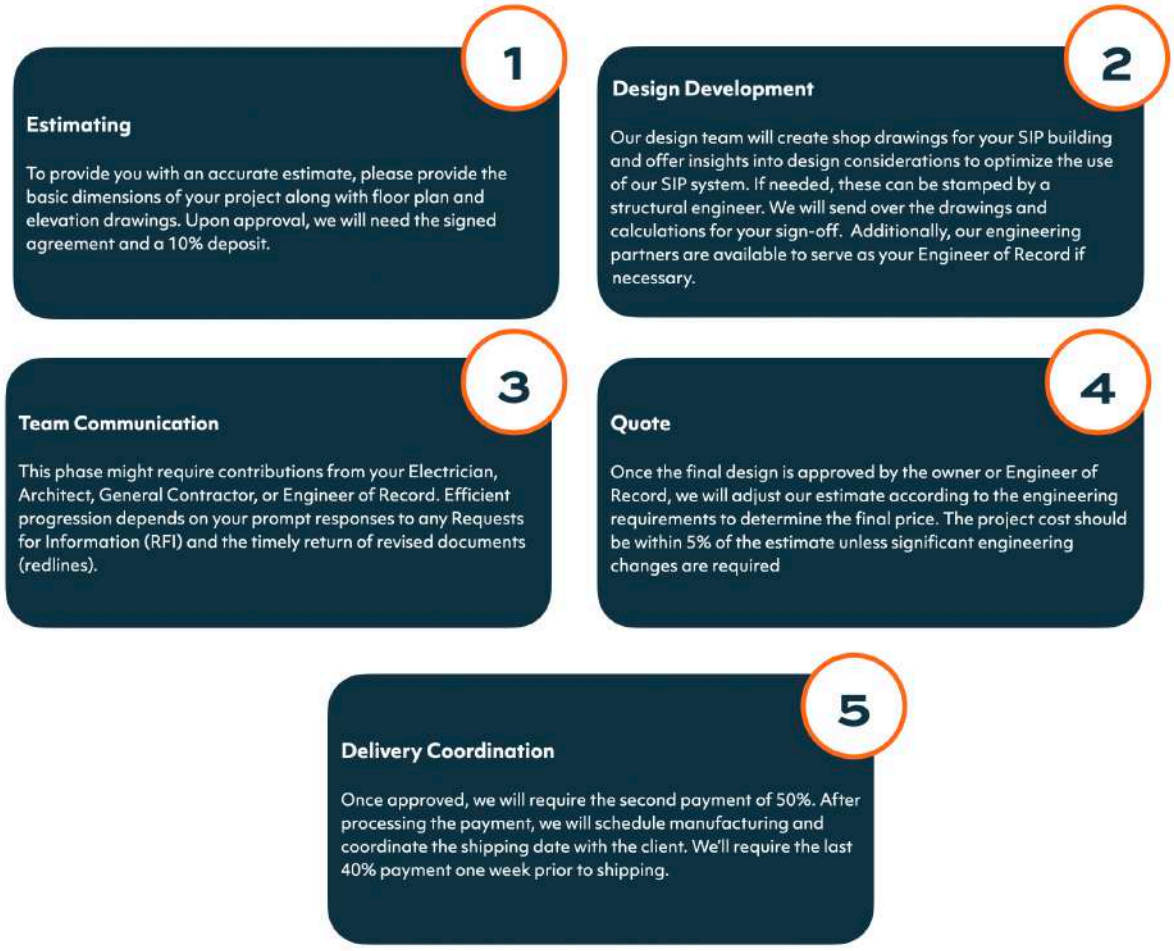
Here is a streamlined overview of our process:

1. 1. **Design and Collaboration:** We start by working closely with you, and if necessary, any third-party stakeholders, to meticulously draft the SIP Layout Drawings. This step ensures that the design meets your specific project requirements.

2. 2. **Finalization and Production Scheduling:** Once the layout drawings are finalized, we schedule production. This stage involves aligning every detail with the precise needs of your project.

3. 3. **Pricing and Production Slot Reservation:** The cost of SIPs is determined by the design complexity, engineering requirements, and fluctuations in market prices. It's important to note that production slots are confirmed only after reaching step 5, due to the potential variability in the timeline of the drawing process.

This structured approach ensures precision in design and efficiency in execution, providing you with a superior construction experience.



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At Preflex, we strive to accommodate client requests for SIP sequencing during shipping while optimizing space utilization and ensuring safe transport. All Preflex SIPs are clearly marked for easy identification. Should you require a specific loading plan, this must be arranged alongside the return of the signed SIP Layout Drawings and may incur an additional fee due to impacts on yield and the total number of shipments needed. Preflex SIPs are produced in 4' and 8' widths and can extend up to 24' in length. Handling of 8' wide SIPs requires 6' forks. For further details, refer to the contract of sale or request a delivery information sheet from PBS.

Please note, as a materials-only supplier, PBS does not take responsibility for errors in design, engineering, or measurement. It is the responsibility of the owner/agent (architect, contractor, and/or installer) to verify all dimensions and specifications. By signing the plans, the owner acknowledges responsibility for their accuracy and agrees to assemble the structure in accordance with the Preflex SIPs Resource Manual.

SIPs Delivery Preparation

1. Familiarize yourself with the SIP installation process, ensuring all necessary materials and hardware are ready. Regional representatives may offer on-site support to guide best practices.
2. Thoroughly review this manual, alongside any project-specific SIP layout drawings, construction details, and notes provided.
3. Verify accessibility of all required framing items needed for installation.
4. For any clarifications post-SIP layout drawing completion, contact your Preflex representative.
5. Arrange for a material handler to assist with offloading and staging SIPs at the construction site.

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SIP Installation Tool Recommendation

- Dunnage for supporting panels (storage off ground, set on level surface, max 4' on center)
- Come-along with 2" trucking ratchet straps (pull panels together)
- Pry bars
- Chalk line
- Lifting plates for roof & tall wall panels
- Framer's square
- Levels (6' or longer)
- Loose 8d and 16d sinker nails (specific to nail gun used)
- Air compressor air hose and electrical cords as needed
- Ladders - step & extension scaffolding for roof panel install if needed
- 1/2" drill for 1-1/2" diameter electrical chase holes & long panel screws
- 1 1/2" auger, forstner or chipper bits (for electrical chase through plates)
- 3/8" drill or impact driver (cordless)
- Chain saw with 14"-16" bar and chainsaw guide for site fabrication
- Power planer
- Recess cutter and/or hot knife
- T25 & T30 Torx driver bit
- Nail gun(s)
- 20oz sealant guns for sausage packs (recommend cordless electric)
- Reciprocating saw (6" & 12" blades)
- One or two circular saws
- Mineral spirits for clean up of caulking gun
- Minimal expanding foam/foam gun compatible with EPS insulation for mechanical openings, window/door jams
- Eye and Ear protection - Fall arrest gear for roofs (if applicable)
- Forklift Notes: 6' Forks are required for 8' wide SIPs

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Installation Best Practices

1. **SIP Handling:** Treat SIPs with care, especially the corners. Avoid lifting from the top edge.
2. **Storage Requirements:** Elevate SIPs and accessories at least 3 inches off the ground. Store flat on supports spaced no more than four feet apart.
3. **Weather Protection:** Cover SIPs with breathable, non-transparent material to protect from weather. Avoid clear or colored plastic.
4. **Fastener Installation:** Ensure fasteners are flush with the SIP surface without overdriving.
5. **Field Cutting Protocol:** Ensure cuts at opening edges stop at the common corner to maintain structural integrity.
6. **Foundation Preparation:** Ensure foundations are level and clear debris from sills before installation.
7. **Installation Accuracy:** Adhere to approved drawings and verify SIP dimensions and electrical chase positions before installation.
8. **Sealing Details:** Follow specified details for applying SIP tape and sealant.
9. **During Installation:** Ensure SIPs are adequately braced.
10. **Nailing Specifications:** Adhere to specific nailing requirements and adjust nail guns for flush finishes.
11. **Electrical Installation:** Utilize pre-formed chases for electrical installations; avoid cutting SIP facings.
12. **Drilling Requirements:** Pre-drill for electrical installations in alignment with chases.
13. **Handling SIPs:** Use appropriate lifting equipment for heavy SIPs.
14. **Spline Usage:** Use continuous structural splines between supports.
15. **Roof Support:** Ensure adequate bearing for roof SIPs per construction details.
16. **Interior Sealing:** Apply interior SIP tape post-roofing.
17. **Roof Moisture Content:** Ensure OSB moisture levels are within recommended limits before covering.
18. **Surface Preparation:** Clean and dry SIPs before applying any materials.
19. **Structural Integrity:** Have SIP roof penetrations reviewed by a licensed engineer.
20. **Flashing and Coverings:** Use code-approved flashings and coverings.
21. **Thermal Barriers:** Apply recognized thermal barriers as per building codes.
22. **Plumbing Installations:** Avoid embedding plumbing within SIPs.
23. **Foam Filling:** Use compatible expanding foam for voids.
24. **Engineering Review:** Have the SIP structure assessed by a licensed engineer.
25. **Detail Priority:** Engineered details supersede generic ones.
26. **Code Compliance:** Ensure the project meets local building codes.
27. **Field Modifications:** Have any SIP modifications reviewed by a structural engineer.
28. **Facing Modifications:** Consult with your Preflex representative before modifying SIP facings; report any inadvertent cuts for structural assessment.

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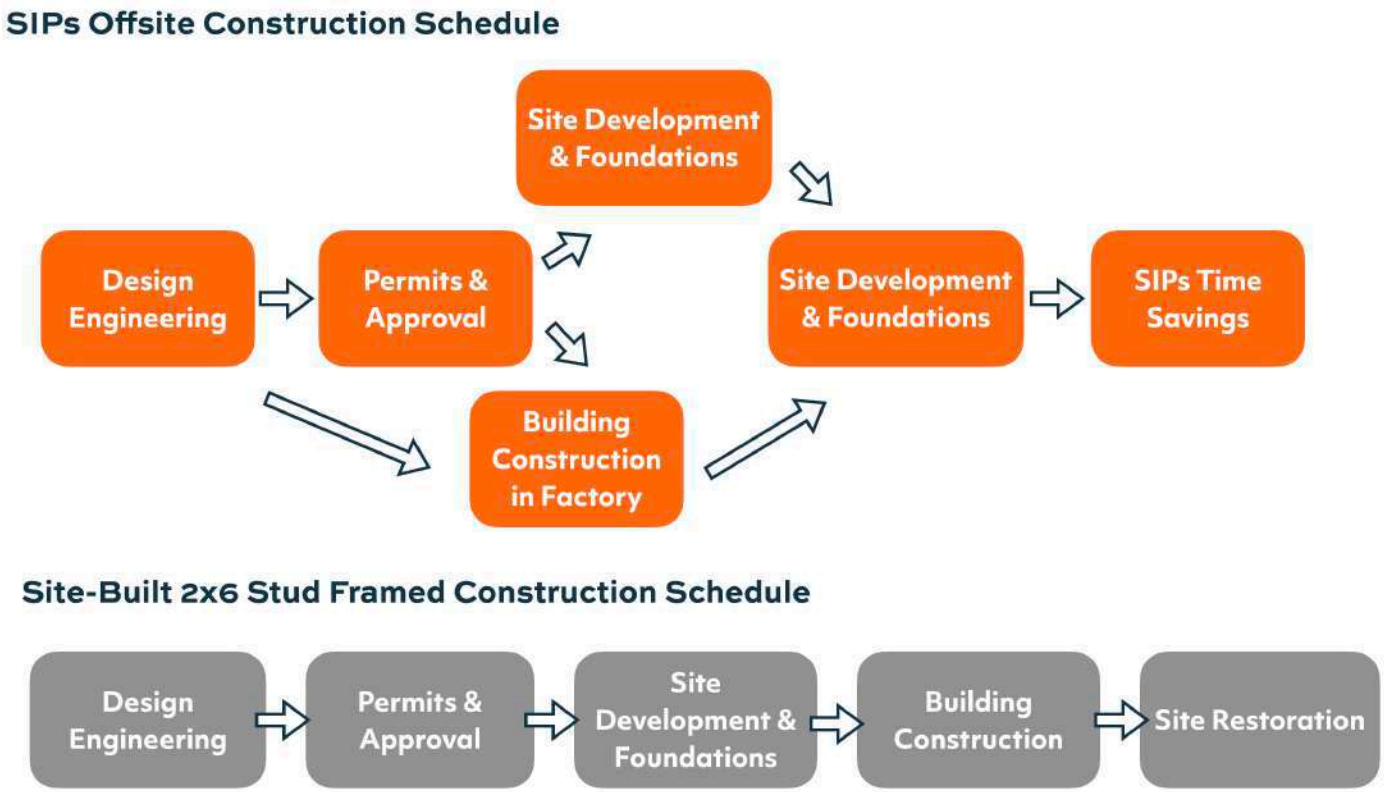
Preflex Project Lifecycle

On behalf of Preflex Building Systems, we thank you for trusting in our innovative SIPs—considered the future of building by many industry experts. Our SIPs are crafted with unmatched precision in our advanced manufacturing facilities, ensuring each panel is tailored to specific project needs. This meticulous approach requires strategic pre-planning, streamlining the on-site assembly process. This guide will navigate you through the project lifecycle, highlighting what to expect at each phase. We encourage you to consult this guide regularly as we progress through your project's milestones with Preflex SIPs.



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The Preflex Process

1. Plan Submission for Estimate:
Submit your building plans, ideally in PDF format, to your local Preflex representative to receive a project estimate within 2-5 business days.

2. Estimate Approval:
Review and approve the estimate, which includes detailed terms and conditions. A signed estimate and a10% deposit are required to proceed.

3. Preflex Point of Contact:
A dedicated Preflex Project Manager (PM) will be assigned to oversee your project, providing SIP layout drawings for your review and approval.

4. Layout Drawing Review:
Thoroughly review the provided SIP layout drawings, which are critical to the customization of your SIPs. Approval of these drawings is essential to move forward.

5. SIP Engineering:
If an engineered stamp is required, our team will collaborate with our partner engineer to have the drawings stamped and the calculations generated. If required, our engineers can serve as the Engineer of Record (EOR) for the project. Any arrangements involving the EOR will be contracted directly with the engineer.

6. Revision Process:
Expect possible revisions to your drawings. Communicate any changes through marked-up drawings to ensure accuracy and avoid delays.

7. Final Drawing Approval:
Once the final SIP layout drawings are approved, they will be converted into production-ready drawings.

8. Approval for Manufacturing:
Once the drawings are approved, we will make any necessary adjustments to the estimate and send the final quote. It should be within 5% of the initial estimate unless significant engineering or design changes were made. We will then issue the second invoice for 50% to schedule manufacturing and delivery

9. Manufacturing:
Your SIPs enter the production queue once all documentation and deposits are finalized. Changes at this stage can affect delivery times and incur additional costs.

10. Final Payment:
We request the final payment of 40% to be settled one week prior to the scheduled shipping date..

11. Receiving Your SIPs:
Prepare to receive your shipment. An on-site representative should be present to inspect and inventory the delivered SIPs. Any discrepancies should be reported immediately.

Project timelines generally range from 4 to 12 weeks, though variations may occur due to multiple factors. As our valued client, you have considerable influence over critical aspects like permitting, engineering modifications, and the payment process. Please be aware that delays in these areas can potentially extend the timeline.

At Preflex, we are committed to delivering top-quality building solutions tailored to your specific needs and timelines, ensuring they meet your high standards. Your investment in Preflex SIPs reflects your dedication to excellence in construction. We take great pride in supporting your projects and encourage you to contact us with any questions you may have at any point during the construction process.

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Thermal Resistance and Thermal Transmittance for the Preflex SIP System with EPS Insulation

Preflex's SIP (Structural Insulating Panel) System is designed for energy-efficient construction, featuring an expanded polystyrene (EPS) insulation core bonded to oriented strand board (OSB) on both sides. This EPS meets the ASTM C578, Type I standard, ensuring high thermal properties and adherence to stringent quality guidelines.

Thermal Performance Metrics:

- **Thermal Resistance (R-value/RSI-value)** and **Thermal Transmittance (U-factor)** are measured at a mean temperature of 75 °F (24 °C), focusing solely on the SIPs without considering additional components like panel framing or cladding.

Key Characteristics of EPS Insulation:

- **Heat Transfer Mechanisms:** Involves minimal convection due to small cell sizes, limited radiation with denser materials, and dominant conduction through the gas within its cells.
- **Durable Thermal Resistance:** The closed cell structure of EPS contains only air, which prevents the decline of thermal resistance over time, unlike other insulations that may experience thermal drift due to gas diffusion.

Enhanced Building Efficiency:

- Preflex SIPs minimize thermal bridging by reducing the need for framing members, which contributes to higher overall R-values. The system also features exceptionally low air leakage rates, drastically reducing energy loss and leading to considerable energy savings in the long run compared to traditional wood frame construction.

By utilizing Preflex SIPs, buildings benefit from improved energy efficiency and significant long-term savings, underscoring Preflex's commitment to innovative and sustainable building solutions.

SIP Total Thickness		Thickness EPS Core & OSB		SIP R-Value @		SIP U-Factor @		SIP Weight	
				75°F	24°C	75°F	24°C		
Inches	mm	Inches	mm	ft2•h•°F BTU	m2•°C W	BTU ft2•h•°F	W m2•°C	lb/ft2	kg/m2
4 1/2	114	7/16 3 5/8 7/16	11 92 11	13.9	2.4	0.072	0.408	3.3	16
6 1/2	165	7/16 5 5/8 7/16	11 143 11	21.1	3.7	0.047	0.267	3.5	17
8 1/4	210	7/16 7 3/8 7/16	11 187 11	27.4	4.8	0.036	0.204	3.7	18
10 1/4	260	7/16 9 3/8 7/16	11 238 11	34.6	6.1	0.029	0.164	3.9	19
12 1/4	311	7/16 11 3/8 7/16	11 289 11	41.8	7.3	0.024	0.136	4.1	20

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Real-World R-Values and the Effectiveness of SIPs

Challenges with Traditional Insulation:

Insulation is essential for energy efficiency in buildings, significantly affecting heating and cooling which constitutes about 50% of home energy use. Traditional fiberglass insulation, commonly used in wood-framed homes, often loses more than half of its laboratory R-value due to factors like thermal bridging and air infiltration. Thermal bridging occurs when heat passes through less insulated materials like wooden studs, reducing the overall effectiveness of the insulation. Additionally, improper installation can leave gaps around windows and wiring, further decreasing its effectiveness.

Advantages of Structural Insulated Panels (SIPs):

SIPs offer a superior alternative, providing both high insulation and structural support with minimal thermal bridging. The Department of Energy's Oak Ridge National Laboratory (ORNL) introduced whole wall R-value testing, which evaluates the thermal resistance of entire wall sections rather than just the insulation material. In these tests, traditional walls with R-19 fiberglass typically perform at R-13.7, while SIPs maintain their full R-value, consistently outperforming fiberglass in real-world conditions.

Impact of Air Infiltration:

Air infiltration through gaps in a home's structure accounts for approximately 40% of heating or cooling loss. SIPs have proven to be significantly more airtight compared to wood-framed homes, reducing the risk of moisture-related issues such as mold growth inside wall cavities and improving indoor air quality.

Laboratory vs. Real World Performance:

Laboratory tests like the guarded hot plate test, which measure the R-value of insulation under ideal conditions, often do not reflect actual performance. ORNL's research found that the R-value of fiberglass insulation decreases in colder temperatures, for example, R-19 insulation performed as R-9.2 at -8°F. In contrast, rigid foam insulation used in SIPs performs better in colder conditions, with expanded polystyrene improving its R-value from R-3.9 per inch at 75°F to R-4.4 per inch at 25°F. Unlike fiberglass, SIPs do not permit convective currents, ensuring more stable and effective insulation across various temperatures.

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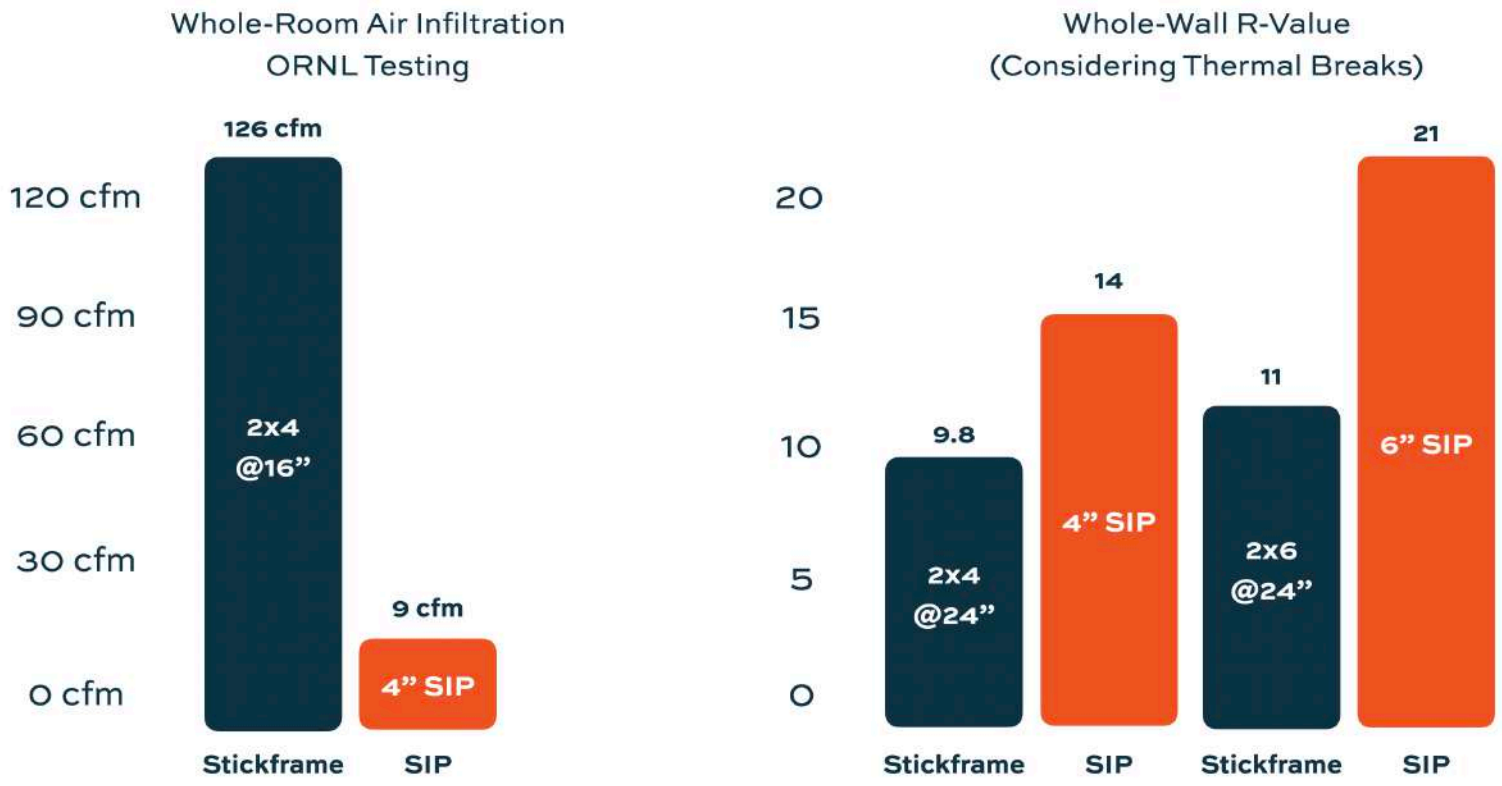
SIP Thermal Performance with EPS Cores

SIP Total Thickness (EPS Type I)	R-value @ 75°F	U-factor @ 75°F	R-value @ 40°F	U-factor @ 40°F	R-value @ 25°F	U-factor @ 25°F	R-value @ 0°F	U-factor @ 0°F
4-1/2-in.	13.9	0.072	15.3	0.065	15.3	0.065	17.1	0.058
6-1/2-in.	21.1	0.047	23.3	0.043	23.3	0.043	26.1	0.038
8-1/4-in.	27.4	0.036	30.3	0.033	30.3	0.033	34.0	0.029
10-1/4-in.	34.6	0.029	38.3	0.026	38.3	0.026	43.0	0.023
12-1/4-in.	41.8	0.024	46.3	0.022	46.3	0.022	52.0	0.019

The Effect Of Air Infiltration



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Preflex Nailbase Panels - Thermal Resistance Values

Preflex Nailbase panels provide a convenient one-step solution for adding insulation along with a nailable exterior surface. These panels consist of 7/16" oriented strand board (OSB) bonded to expanded polystyrene (EPS) rigid insulation.

A common application for Preflex Nailbase panels is atop structural roof decks, offering an insulated roof system with a surface suitable for nailing. These panels are versatile and suitable for use in both residential and commercial construction projects.

Preflex Nailbase Panel Benefits:

- Provides continuous insulation over a structural support system, minimizing energy loss by eliminating thermal breaks.
- Features a nailable surface, allowing for easy attachment of exterior cladding.
- Offered in various insulation thicknesses and panel sizes to suit different project requirements.

Preflex Nailbase Panel Thermal Resistance

Panel Thickness	Components		Thermal Resistance	
	Insulation Thickness	OSB Thickness	R-Value (ft2•hr•°F/BTU)	RSI (m2•°C/W)
4-1/16"	3-5/8"	7/16"	14.8	2.61
6-1/16"	5-5/8"	7/16"	22.6	3.98
7-7/8"	7-3/8"	7/16"	28.5	5.02
9-7/8"	9-3/8"	7/16"	37.0	6.52
11-7/8"	11-3/8"	7/16"	44.8	7.89

Thermal resistance values at mean temperature of 75 F (24 C) for Preflex Nailbase panel only.

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OSB Frequently Asked Questions

What is OSB?
Oriented Strand Board (OSB) is a versatile, high-performance structural panel used widely in construction, packaging, furniture, and DIY projects. Known for its uniformity, strength, and versatility, OSB can be customized in size, thickness, texture, and strength to meet specific requirements.

How is OSB manufactured?
OSB is made from sustainably harvested trees such as aspen poplar and southern yellow pine. The manufacturing process involves cutting and debarking logs, forming them into precise strands, which are then mixed with wax and a waterproof binder, layered cross-directionally, and compressed under high heat and pressure to ensure optimal strength and durability.

What are the performance guarantees and standards for OSB?
OSB panels are guaranteed to meet high standards set by bodies such as the CSA, PS2, EN, and JAS. These panels come with performance guarantees ensuring they meet specified criteria when installed properly. If they fail to meet these standards, they will be replaced or refunded by the manufacturer.

How does OSB compare to other structural panels?
OSB is comparable to other structural panels in terms of strength, rigidity, size, and fastener holding capabilities. It meets building code requirements and is rated Exposure 1 for durability, making it suitable for use during construction delays without special treatment.

Is OSB cost-effective and environmentally safe?
OSB is economically produced using globally sourced, sustainable materials, making it cost-effective and reliable. It is environmentally friendly, made from renewable wood resources, and recyclable into other products. OSB panels emit minimal or no formaldehyde, adhering to strict environmental and health standards.

What are the specific environmental considerations and safety features of OSB?
OSB panels are safe for installation and disposal. They feature a textured surface for safe handling during roofing installations and are considered safe for landfill disposal due to their natural wood content and minimal additives. Modern mills producing OSB adhere to stringent environmental standards to minimize ecological impact.

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Pest Infestation Management Strategies Overview

This bulletin provides guidelines on effective building practices to prevent infestations of carpenter ants and termites, emphasizing prevention during construction as a more manageable and cost-effective approach than addressing an infestation after completion.

Building Practices for Prevention
Adhering to these building practices is essential for minimizing the risk of pest infestation:

Clearing and Land Management:
- Remove stumps, roots, and wood debris around the building to maintain a buffer zone of about 30 meters (100 feet).
- Trim back overhanging tree limbs to prevent aerial access by pests.
- Regularly inspect and seal potential entry points to the structure.

Material Management:
- Avoid using natural nesting materials like dead trees near the construction site.
- Store firewood away from the house to prevent easy access for pests.

Chemical and Physical Barriers:
- In pest-prone areas, particularly those that are heavily wooded or damp, use baiting systems or perimeter treatments with long-lasting pesticides before backfilling.
- Treat the soil around the building post-backfilling and install termite shields under the sill in termite-prone zones.
- During construction, treat wooden components with borate-based compounds, effective against both carpenter ants and termites and safe for humans.

Landscaping and Structural Practices:
- Promote proper drainage away from the foundation to discourage pest habitation.
- Maintain necessary clearances: approximately 300 mm (12 inches) to the underside of the base plate or framing and 150 mm (6 inches) to the siding.

Additional Resources and Compliance

For more detailed guidance on integrated pest management strategies, relevant resources include:

- The American Wood Council's "Design of Wood Structures for Permanence" (Wood Construction Data No. 6)
- The Canadian Wood Council's "Termite Control and Wood-Frame Buildings"

Always consult local building codes for region-specific guidelines and consider engaging a local pest control expert for customized pest management solutions.

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Heating, Ventilation and Air Conditioning (HVAC) System Requirements

This bulletin outlines HVAC system design requirements for energy-efficient buildings constructed with the Preflex Structural Insulated Panel (SIP) System.

Preflex SIP System Warranty
The Preflex Inc. warranty specifies that it does not cover damage resulting from inadequate design, performance, or maintenance of the heating, ventilation, and air conditioning (HVAC) system. Building owners are responsible for ensuring proper design and maintenance of the building HVAC system.

Design Considerations for Heating and Air Conditioning System
The Preflex SIP System creates an energy-efficient building envelope with high thermal resistance and reduced air leakage. This often leads to savings in the size of heating and air conditioning equipment needed for a building. HVAC system contractors must account for the higher effective R-value and increased air tightness provided by the Preflex SIP System in their design calculations, resulting in significantly reduced HVAC heating and cooling requirements.
Proper sizing of the HVAC system is critical to avoid inefficiencies. Oversized systems fail to reach steady operating rates, leading to less energy efficiency and increased maintenance compared to properly sized equipment.

Ensuring Adequate Ventilation
Buildings constructed with the Preflex SIP System are typically highly airtight, necessitating controlled ventilation to exhaust fumes from indoor sources and regulate interior relative humidity. Ventilation involves removing stale air and excess moisture from occupied areas while introducing outdoor air to dilute indoor air contaminants. It's important to differentiate ventilation from air filtration and air cleaning.

Only systems that either remove air from the building or supply outside air to the building are categorized as ventilation systems. Mechanical ventilation ensures reliable control over air exchange, enhances indoor air quality under most circumstances, except when outdoor air quality is poorer than indoors, and reduces indoor relative humidity levels in winter, thereby minimizing surface condensation on cold surfaces like windows.

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Types of Mechanical Ventilation:

- **Supply-only systems:** These systems solely supply outdoor ventilation air mechanically without mechanical exhaust. An example is a naturally aspirated furnace with an outdoor air supply into the return air plenum. However, this isn't typically recommended for airtight buildings due to the risk of building pressurization leading to excessive air leakage and potential moisture damage.

- **Exhaust-only systems:** In exhaust-only ventilation, exhaust is mechanically provided, while supply air enters through cracks, holes in the building envelope, or passive wall diffusers. Common examples include buildings with kitchen and bath fans or central exhaust ventilators. These systems often lack control over ventilation air supply and distribution and cannot filter incoming air. They may also cause back venting of combustion appliances and entry of soil gases without preventive measures.

- **Balanced Ventilation Systems (Heat Recovery Ventilators):** Balanced ventilation systems supply fresh air and exhaust stale air at equal rates by using fans to move both air streams. These systems typically incorporate a heat recovery ventilator (HRV), which contains one or two fans and a heat-exchange core. One fan extracts stale air, passing it through the heat exchange core, while the other fan brings in outside air through the same core. The two air streams don't mix directly, but the incoming air stream is warmed by the heat from the outgoing air stream.

Regardless of the chosen ventilation system, the design professional overseeing the building's overall design should be directly involved in HVAC system design. Additional information on HVAC system design for residential applications can be found through sources such as the US Department of Energy – Energy Efficiency and Renewable Energy.

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Roof Ventilation Building Code Standards

The Preflex SIP system is a closed cavity building component that differs from the concept of "enclosed rafters" as defined in building codes. Preflex SIPs are comprised of an expanded polystyrene (EPS) insulation core material with structural grade 7/16" oriented strand board (OSB) factory laminated to both the top and bottom surfaces of the rigid EPS core.

In traditional wood frame construction, an enclosed rafter space is created when interior ceiling finishing is directly applied to the underside of rafters, with exterior roof sheathing installed on the top side of rafters. Insulation material is inserted into this "enclosed" space to establish an insulated cathedral ceiling. Ventilation requirements outlined in US and Canadian codes ensure that if warm air infiltrates this enclosed space and leads to moisture condensation on the cold top side of rafters or the underside of roof sheathing, it has the opportunity to evaporate.
However, due to the Preflex SIP system's design, where the EPS insulation is in direct contact with the entire underside/interior of the structural roof deck (top skin of the SIP), condensation within a Preflex SIP roof assembly is unlikely. Additionally, joints between Preflex SIPs are sealed to prevent any air movement into the spaces between panels, further mitigating the risk of moisture buildup.

Preflex SIP roof joint sealing details incorporate two layers of redundancy to prevent any air movement within the joint:

- Joint sealant is applied to the vertical face, as well as the top and bottom face of the 2x wood spline used for joining the SIPs.

- Panel seal tape is applied to the underside of SIP roof joints. Additionally, a sealant is applied to the top edge of OSB skins to seal the top surface of the SIP joint.

Given that there is no space within the SIP nor panel joints for air movement to occur, the Preflex SIP System adopts a closed cavity design. Consequently, the area beneath the SIP roof constitutes fully conditioned space, eliminating any opportunity for condensation to develop.

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LEED v4 New Construction (revised 2017) Summary

This LEED section highlights how using SIPs in construction can significantly contribute to earning LEED points across several categories by enhancing energy performance, using responsibly sourced materials, and improving indoor environmental quality.

EA – Energy and Atmosphere:

- **EA 2: Optimize Energy Performance – Maximum 18 Points**

Projects can earn up to 18 points for energy savings verified by whole building energy modeling per ANSI/ASHRAE/IESNA Standard 90.1-2010, Appendix G or by meeting prescriptive standards for commercial buildings. SIPs (Structural Insulated Panels) contribute to these savings by reducing energy needed for heating and cooling, which constitutes 44% of energy use in commercial buildings.

MR – Materials and Resources:

- **MR 3: Building Product Disclosure and Optimization – Sourcing of Raw Materials (Option 2) – 1 Point**

One point is awarded if at least 25% of the total value of permanently installed building products are sourced from responsibly extracted materials, such as FSC-certified OSB, which is available from some SIP manufacturers.

IEQ – Indoor Environmental Quality:

- **IEQ 2: Low Emitting Materials – 1-3 Points**

SIPs qualify for 1-3 points since the structural adhesives and OSB used meet the low emitting materials requirements and do not contain added urea-formaldehyde resins.

- **IEQ 3: Construction Indoor Air Quality Management Plan – 1 Point**

As low-VOC products, SIPs help maintain required VOC levels for healthy indoor air quality.

- **IEQ 5: Thermal Comfort—Design – 1 Point**

The airtight and well-insulated properties of SIPs contribute to thermal comfort, earning another point.

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LEED for Homes Summary

This summary outlines how homes using SIPs can achieve high LEED points across various categories by optimizing energy performance, using environmentally preferable products, and implementing efficient construction practices.

EA – Energy and Atmosphere:

- **EA 1: Annual Energy Use – Maximum 29 Points**

Homes can earn up to 29 points under the Performance Path by demonstrating energy efficiency through a HERS Index. This index considers insulation quality, HVAC efficiency, and other factors. Points range from 0 (ENERGY STAR level) to 29 (net-zero energy). ENERGY STAR compliance is a prerequisite.

- **EA 7: Air Infiltration – Maximum 2 Points**

SIP homes can earn up to 2 points for air tightness, assessed through blower door tests.

- **EA 8: Envelope Insulation – Maximum 2 Points**

Homes using SIPs can easily exceed the 2004 IECC insulation standards by 5%, validated via ENERGY STAR SIP Visual Inspection Form.

MR – Materials and Resources:

- **MR 2 Environmentally Preferable Products – Maximum 4 Points**

Using FSC-certified OSB in SIPs grants 0.5 point per component (walls, roofs, floors), with an additional 0.5 point per component if manufactured within 100 miles.

- **MR 3 Construction Waste Management – Maximum 3 Points**

SIPs reduce on-site waste, with points allocated based on the efficiency of waste reduction per square foot, ranging from 0.5 to 3 points.

- **MR 4 Material-Efficient Framing – Maximum 2 Points**

Each SIP system used in walls, roofs, and floors can earn 0.5 to 1 point. Alternative material-efficient framing techniques are also eligible for points.

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Guidelines for Design of Roof Cladding

The Preflex SIP system requires protection from rain exposure before, during, and after installation to ensure long-term performance. Continuous protection from water penetration is crucial for the projected lifespan of the structure.

The oriented strand board (OSB) used in the Preflex SIP System meets standards such as US DOC PS2-04, Exposure 1, for exterior grade sheathing. This classification indicates suitability for uses not permanently exposed to weather, with limited exposure to moisture.

Despite its strengths, OSB tends to swell when exposed to moisture, especially at the edges. Installing a roof sheathing membrane over the SIP can temporarily shield the OSB from moisture if shingles are applied shortly after. However, applying the membrane or shingles while the roof deck is damp can slow moisture transfer and lead to OSB edge swell. It's crucial to allow wet OSB to dry thoroughly before installing the membrane or shingles. Any OSB edge swelling at SIP joints should be flattened before membrane installation to prevent telegraphing through the shingles.

In conventional wood frame structures, water leakage into the attic alerts occupants to roof cladding failures. However, in closed panels like the Preflex SIP system, occupants may not notice water penetration until the exterior OSB skins absorb excessive moisture.

To address this, the roof cladding design must incorporate a secondary defense against exterior water penetration, beyond asphalt shingles. This second line of defense should align with building code requirements and geographical conditions, potentially involving modified bituminous sheathing membranes.

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2009 International Residential Code, Section R613: Construction of Structural Insulated Panel Walls

Section R613 of the 2009 International Residential Code® recognizes the use of Structural Insulated Panels (SIPs) in residential construction applications. Under this section, project drawings, typical details, and specifications are not required to bear the seal of the architect or engineer responsible for design, unless mandated by state law.

R613 specifically addresses the construction of exterior SIP walls and interior load-bearing SIP walls for buildings not exceeding certain dimensions. These restrictions include a maximum length of 60 feet perpendicular to the joist or truss span, a maximum width of 40 feet parallel to the joist span or truss, and a maximum height of two stories with each story not exceeding 10 feet in height.

Exterior walls constructed in accordance with R613 are considered load-bearing, and SIP walls must adhere to specific environmental and structural limitations. These limitations include maximum wind speed, ground snow load, and seismic zones.
The IRC provision's scope was determined through a test program conducted by APA, The Engineered Wood Association.

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Sheathing Membrane (Water-Resistive Barrier) Requirements

The purpose of this information bulletin is to outline the specifications for sheathing membranes, also known as weather-resistive barriers, utilized in wall assemblies constructed with the Preflex SIP System.

Exterior finish materials commonly allow water infiltration from outside elements, particularly during periods of heavy wind and rain. The role of a sheathing membrane in above-grade wall applications is to act as a secondary barrier, preventing exposure of the Preflex SIP System's exterior OSB skin to liquid water, such as rainwater. While sheathing membranes impede bulk water movement into the wall assembly, they are typically designed to permit the passage of smaller moisture vapor molecules.

A separate sheathing membrane meeting the following requirements must be promptly applied over the external surface of the Preflex SIP System after installation. It is crucial that the sheathing membrane be installed beneath exterior cladding to facilitate water drainage to the exterior. Additionally, appropriate flashings must be utilized at door, window, and other penetrations in the building envelope.

Various breather-type polyolefin sheathing membranes are available for wrapping the building envelope exterio. Alternatively, the required sheathing membrane can be provided using traditional asphalt-impregnated building paper meeting ASTM D226 standards for Type 1 No. 15 asphalt felt.

In regions susceptible to wind-driven rain, it is strongly recommended to incorporate a minimum 3/8-inch-wide airspace into the wall system between the exterior cladding and the sheathing membrane. This airspace forms a drainage cavity between the exterior cladding and the sheathing membrane, aiding in water drainage. For siding materials, this can be achieved by vertically strapping or furring out the walls after applying the sheathing membrane. Ventilation at the top and bottom of the airspace is also advised to facilitate drying of any moisture accumulation on the weather-resistive barrier surface.

It is advisable to consult local building code requirements for additional specifications, as local regulations may impose more stringent requirements than those listed above.

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Tools and Materials

In this bulletin we highlight essential tools and materials designed to enhance the efficiency and quality of your construction projects. We feature a range of innovative products which we work with, from versatile fasteners to advanced sealing technologies, all engineered to meet the stringent demands of Preflex and the SIPs building environments.

OMG HeadLok

The OMG HeadLok is a high-performance, versatile flat head fastener ideal for a variety of panel applications including Structural Insulated Panels, Prefabricated Wall Panels, Nailboard, and more. It is compatible with wood, structural concrete, purlins, corrugated, and structural steel substrates. For optimal installation, utilize a high torque, low RPM screw gun, ensuring the washer head remains flush with the surface without overdriving. The fastener requires specific point styles for different substrates, such as spade points for structural concrete to ensure a minimum 1-inch penetration and drill points for purlin attachment. A mandatory pullout test is recommended before project commencement to evaluate the fastener's performance.

SIP Seal Vapor Tape Ex

SIP Seal Tape Ex is an all-weather, durable joint sealant crafted for SIPs and panelized construction. This sealant integrates a hybrid synthetic rubber formula that adheres strongly to OSB and engineered wood. Offering high initial tack, this tape is thermally stable and smoothly conforms to irregular surfaces. It acts as an air and vapor sealing tape compatible with a variety of construction materials including OSB, plywood, and EPS foam. The sealant is 100% solid, contains no VOCs, and provides UV stability up to 6 months while being hydrophobic and mold-resistant, ensuring a long-lasting, vapor-tight seal.

IRC 2021 Vapor Retarder Standards

The IRC 2021 emphasizes the importance of moisture control in residential buildings by mandating the installation of vapor retarders on the warm-in-winter side of insulation in framed walls, floors, and roof/ceilings that are part of the building's thermal envelope. These materials must have a permeance rating of 1 perm or less. The Preflex SIP System meets these standards with a tested composite vapor permeance of 0.5 perm, and ensures proper sealing and continuity of the vapor barrier through specific installation techniques.

SIP Seal Joint Sealant

This robust, industrial-grade polyether sealant offers exceptional flexibility with up to 1,200% elongation while maintaining its integrity. It is engineered to cure in the presence of moisture and withstands extreme temperatures ranging from -75°F to 300°F. The SIP Seal Joint Sealant adheres well to a wide range of materials and is particularly suited for joints with high movement potential. Packaged in 20-oz sausage packs, it features non-sagging properties, a rapid curing time, and meets or exceeds multiple ASTM standards, making it a versatile choice for both indoor and outdoor applications.

Each product in this bulletin represents a commitment to quality and efficiency, designed to meet the diverse needs of modern construction projects.

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Preflex SIP System – Low Slope Roofing Applications

Roofing systems utilizing single-ply elastomeric, thermoplastic, or polymer-modified bitumen membranes are compatible with the Preflex SIP System for constructing low-slope roofs on residential or commercial structures. It is recommended to maintain a minimum roof slope of 2% or ¼" per foot (20 mm per meter) to ensure proper drainage.

Single-ply membranes can be installed in various methods: loose-laid and ballasted, partially or fully adhered, or mechanically fastened. When opting for loose-laid and ballasted membranes over a Preflex SIP roof deck, a slip sheet should precede the roof membrane to serve as a separation layer. For adhered or mechanically fastened membranes, a protection board like gypsum board or wood fiberboard may serve as the separation layer.

The inclusion of a separation layer helps minimize potential damage to the Preflex SIP in case future removal of the roof membrane becomes necessary. Furthermore, it's imperative to prevent adhesives or solvents used for sealing membrane seams from contacting the EPS insulation core within the Preflex SIP. Hence, staggering the joints between protection boards with those of the Preflex SIP is advised. Maintaining the integrity of the Preflex SIP system is crucial as it ensures the structural capabilities it was engineered to provide remain intact.

Designing low-slope roof systems should adhere to the requirements outlined in relevant local building codes. For commercial roof applications, one vital consideration is the classification of exterior roof coverings, typically falling into Class A, B, or C categories as determined by testing methods such as ASTM E108/UL 790 in the United States.

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Understanding New Tax Credits and Benefits of SIPs

It pays to build sustainably, here's a breakdown of how SIPS can help you meet new tax credit requirements.

The largest investment in fighting climate change in U.S. history is here. The 2022 Inflation Reduction Act (IRA) brings a fresh wave of tax credits for energy-efficient buildings and extends the terms for claiming existing credits. It's an exciting new era for the sustainable construction industry. As a builder or developer, you can now take advantage of these incentives to reduce costs and lower your carbon footprint.

Whether it's efficiency improvement to a multifamily housing complex, a new single-family build meeting ENERGY STAR requirements, a commercial retrofit, or a property owner installing new insulation, Structural Insulated Panels (SIPs) make it easy to build sustainably and reap the benefits of the new tax credits. SIPs can help you qualify for incentives & rebates for efficiency upgrades to new and existing commercial buildings, homes, multifamily and public buildings through sections 45L, 25C, and 179D of the IRA.

First, here's a rundown of what IRA tax credits and deductions have been updated and extended…

SEC. 45L HOMEBUILDER TAX CREDIT

Section 45L incentivizes homebuilders and multifamily developers to reduce energy consumption with a per dwelling unit tax credit. Key credit details include:

- New homes certified under the ENERGY STAR programs will be eligible for a $2,500 credit
- The credit is increased to $5,000 for dwelling units certified under the Zero Energy Ready Homes Program (ZER).
- For Multifamily units, the base qualification decreased to $500, while ZER multifamily units are $1,000 each.
- Multifamily units meeting prevailing wage requirements will receive an increased $2,500 credit for base qualification and $5,000 for ZER.
- Initially, low-rise residential developments were the only ones eligible for this credit, but it is now extended to homebuilders and single-family or multifamily developers. The base qualification requirements for these developers are that they must meet both national and local energy-saving requirements.

Meeting these performance levels can be accomplished cost-effectively through proper early ***design with SIPs***, because SIPs meet and exceed building code thermal envelope requirements and eliminate additional continuous insulation needs on the building exterior (meeting the 2021 and 2018 IECC building codes). The 45L tax credit is a great opportunity to reduce project costs, increase your cash flow, reduce maintenance costs, and enhance your ROI. Continue reading to learn how SIPs can help meet the above-detailed requirements. ***Visit the IRS updated Form 8908 for new requirements and qualifications.***

On September 27, 2023, the Internal Revenue Service (IRS) issued ***Notice 2023-65***, which contained ***guidance for taxpayers on the § 45L credit*** as amended by IRA. The Notice establishes that to be eligible for the higher tier credit, a home or apartment must be certified under the applicable ZERH program requirements as defined on the ***DOE ZERH Program Requirements webpage***. DOE has updated the ZERH Program Requirements webpage and the ***Section 45L Tax Credits for Zero Energy Ready Homes*** webpage in coordination with the release of the IRS Notice and guidance. Stakeholders can use these materials to determine which version of ZERH program requirements will apply to their project for 45L eligibility. Individuals or entities seeking to claim the 45L tax credit should consult with a tax professional to determine whether and how they can claim the credit and determine whether the credit can be used with other tax

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incentives or Federal incentives.

SEC. 25C HOME EFFICIENCY IMPROVEMENT TAX CREDIT

Formerly known as the Nonbusiness Energy Property Credit, the IRA extended and revitalized this tax credit by substantially increasing the amount homeowners can claim on their taxes for energy-efficient home improvements such as insulation, exterior doors, and similar energy-saving improvements.

• The expanded 25C will provide a 30% tax credit (up to $1200 annually) to homeowners who take certain steps to make their homes more energy efficient.
• 25C tax credits can also be combined with rebates offered through DOE's forthcoming $8.8 billion Home Energy Rebates Program. However, these funds will not be available to homeowners until at least early 2024.

With retrofit insulated panels you can get a higher R-value and a quicker application than the alternatives, helping you to qualify for insulation credits under 25C. *Learn more about* ***how going green could help taxpayers qualify for expanded home energy tax credits.***

SEC. 179D TAX DEDUCTION FOR ENERGY-EFFICIENT COMMERCIAL BUILDINGS

Expanded tax deductions, equaling up to $5 a square foot, are available for new and existing energy-efficient building projects. The tax deduction is available for new and existing nonprofit, government, and commercial building projects that achieve specific energy efficiency goals and ASHRAE standards. ***Learn more about Sec 179D Tax Deduction and SIPs***

HOW SIPS CAN HELP YOU QUALIFY FOR IRA TAX CREDITS

By building with SIPs, you can easily achieve a high-performance building envelope that can help you qualify for the new tax credits and incentives. SIPs provide a continuous insulation envelope that minimizes thermal bridging and reduces air infiltration, resulting in significantly lower heating and cooling costs. *Learn about the full advantages of designing and building with SIPs through our* ***SIP Design Best Practices Series.***

• SIPs meet and exceed building code thermal envelope requirements and eliminate additional continuous insulation needs on the building exterior (meeting the 2021 and 2018 IECC building codes).
• SIP homes save 40-60% on energy than stick-frame construction, depending on the climate zone.
• SIPs provide extremely airtight structures, a key component in improving indoor air quality (IAQ).
• SIPs provide 85-88% less air leakage than stick-frame construction.
• SIPs are available in a range of thicknesses delivering exceptional thermal performance.
• Various SIP connections are available which minimize thermal bridging.
• Factory-applied rigid foam insulation cores eliminate concerns over quality insulation installment.
• Reduced HVAC requirements.
• Applying retrofitted panels to the exterior of a home can get you higher R-Values.

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BUILD

Building Incentive for Low-Emission Development Program

Program Fact Sheet

The Building Initiative for Low-Emissions Development (BUILD) Program is a residential building decarbonization program that provides incentives and technical assistance to support the adoption of advanced building design and all-electric technologies in new, low-income all-electric homes and multifamily buildings.

300 FREE HOURS **of technical assistance** for design, development, and construction	**$3,300** **per unit for multifamily** average greenhouse gas incentive
$2,000,000 **program cap** per applicant	**$5,500** **per home for single family** average greenhouse gas incentive

New Adopter Design Award

BUILD provides eligible applicants constructing their first all-electric, low-income multifamily building (10+ units) with **up to $100,000 design award** to defray direct design costs.

Incentives

BUILD provides robust incentives consisting of four components:

Base Incentive

Greenhouse Gas Emissions Avoided

Incremental PV Incentive

Flat Rate of PV Above Code

Building Efficiency Incentive

Percentage Above Code

Kicker Incentive

Additional GHG Reduction Technologies

FOR QUESTIONS, EMAIL BUILD AT:

BUILD@energy.ca.gov

FOR MORE INFORMATION, VISIT:

BUILD Program Webpage

www.energy.ca.gov/programs-and-topics/ programs/building-initiative-low-emissions-development-program

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Who is Eligible?

- Building owners or developers of low-income housing with:
- Five years' development experience
- At least one completed low-income project

What Projects are Eligible?

- Single and multifamily low-income residential housing with varying income restrictions depending on project location and type of housing
- All-electric and no hookups to the gas distribution grid
- Demonstrate modeled resident utility cost savings
- Project located in one of four gas service territories: Southern California Gas, Pacific Gas & Electric, San Diego Gas & Electric, and Southwest Gas Corporation

What Residential Buildings are Eligible?

- Newly constructed buildings
- Existing buildings repurposed for housing
- Building additions or renovations where 50% or more of the exterior weight bearing walls are removed or demolished

Why Participate in BUILD?

- Receive up to $2,000,000 in incentives to reduce construction costs
- Reduce long-term utility costs for both you and your residents or home buyers
- Eliminate costs for gas infrastructure
- Layer funds with other programs and BUILD incentives

Free Technical Assistance Services

- Educational resources on all-electric building design and technologies
- Support throughout all development phases (including building design, construction, installing near-zero emission technologies) and information on local building permits
- Assistance with submission of BUILD Incentive Application package and participation support

Get Started Today

buildapp.energy.ca.gov

BUILD is administered by the California Energy Commission in collaboration with the California Public Utilities Commission. It is authorized by Senate Bill 1477 (2018, Stern) and funded by the four California gas corporations apportioned according to each gas corporation's percentage share of allocated Cap-and-Trade Program allowances. Reservations are approved on a first come, first served basis, and regional funding availability is based on the utilities' contribution to the program.

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3.0 DESCRIPTION

3.1 General:
Structural Insulated Panels are factory-assembled, engineered-wood-faced, structural insulated panels (SIPs) with an expanded polystyrene (EPS) foam core. The product is intended for use as load-bearing or non-load- bearing wall and roof panels. Structural Insulated Panels are available in 45/8-inch (117.5 mm) through 15-inch (381 mm) overall thicknesses and are custom-made to the specifications for each use. The maximum product size is 8 feet (2438 mm) in width and up to 24 feet (7315 mm) in length.

3.2 Materials: 3.2.1 Facing: The facing consists of two single-ply oriented strand board (OSB) facings a minimum of 7/16-inch-thick (11.1 mm) conforming to the properties shown in Table 1. Additionally, facing materials shall conform to DOC PS 2, Exposure 1, Rated Sheathing with a span index of 24/16. Panels may be manufactured with the facing strength axis oriented in either direction with respect to the direction of product bending provided the appropriate design values are used.

3.2.2 Core: The core material is EPS foam plastic insulation conforming to ASTM C578, Type I or ASTM C578 Type VIII. The foam core, up to 4-inch (101.6 mm) thickness, has a flame spread rating not exceeding 75 and a smoke-developed rating not exceeding 450 when tested in accordance with ASTM E84. Cores used in structural insulated panels up to 15 inches (381 mm) thick, comply with IBC Section 2603.3 Exception 4.

3.2.3 Adhesive: Facing materials are adhered to the core material using a thin-film adhesive. The adhesive is applied during the lamination process in accordance with the in-plant quality system documentation.

3.2.4 Material Sources: The facing, core and adhesive used in the construction of Structural Insulated Panels must be materials from approved sources as identified in the in-plant quality system documentation. A list of material suppliers is provided in Table 15.
3.2.5 Splines: Structural Insulated Panels are interconnected with surface splines, block splines, or I-joists (Figure 1). Connections using dimensional lumber splines or engineered structural splines not specifically addressed in this report must be designed in accordance with accepted engineering practice to meet applicable code requirements.

3.2.5.1 Surface Splines: Surface splines (Figure 1) consist of 3-inch-wide (76.2 mm) by 7/16-inch-thick (11.1 mm) or thicker OSB. At each panel joint, one surface spline is inserted into each of two tight-fitting slots in the core. The slots in the core are located just inside the facing.

3.2.5.2 Block Splines: Block splines (Figure 1) are manufactured in the same manner as the SIP except with an overall thickness that is 1 inch (25.4 mm) less than the overall thickness of the panels to be joined.

3.2.5.3 I-Joist Splines: Structural capacities for prefabricated wood I-joists splines (Figure 1) shall be established and monitored in accordance with ASTM D5055 with properties equal to or greater than those shown in Table 2. The overall depth of the joist is 1 inch (25.4 mm) less than the overall thickness of the panels to be joined.

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JOINT SEALANT
TECHNICAL DATA SHEET

Product Description

- Industrial grade, highly durable and flexible polyether sealant will provide up to 1,200% elongation while maintaining cohesive integrity
- Ready to use, cures in the presence of moisture
- Will not harden, yellow, or lose its flexibility
- Weather resistant, Type II for extended water contact
- Highly UV resistant (no change at 2,000 UV-A hours)
- Smooth gunning, workable in low temperatures

Applications

- A highly flexible sealant for use in construction joints that have high potential for movement
- Bonds and maintains a flexible seal with OSB, wood, EPS foam, concrete, ceramics, metal, vinyl, PVC plastic, phenolic panels, ABS, and other non-asphaltic substrates. (Always check adhesion).
- Indoor/outdoor use

Package/Colors

- 20-oz sausage pack: Lime (61009)
- 20 count per case
- 45 cases per pallet

Technical Data

- Specific gravity 1.4 -1.44
- Tack-free time 60 mins (77°F, 50% rh)
- Skin Time 30 mins (77°F, 50% rh)
- Non-sagging, non-corrosive, non-staining
- Tensile strength 125 psi
- Viscosity 500,000 cps
- Elongation 1,200%, 100% modulus 20 psi
- Peel adhesion 23 ppi/100% CF
- Hardness Shore A 15
- High-temperature resistance 300°F for brief periods; low-temperature flexibility -75°F

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Environmental

- Intertek Gold Certified - No hazardous air pollutants (HAPS-free)
- 99% solids, no solvents
- No isocyanates
- Low VOC, less than 12-grams/liter
- Shelf life of 18-months from date of manufacture, stored below 80°F

Engineering Standards

- Meets or exceeds ASTM C920, Type S, Grade NS, Class 50. Use NT, M, A, G and O
- UV stability ASTM G26
- USDA compliant

Usage Instructions

- Apply to clean, dry, frost-free surfaces to ensure proper seal
- Apply in continuous beads in accordance with SIP manufacturer's manual
- Application temperature of -20°F to 120°F.
- Use with adequate ventilation inhalation of vapor during application and cure may cause slight eye and throat irritation
- In case of contact with eyes, lips, or mouth flush thoroughly with water. If irritation persists, consult a physician
- Avoid repeated, prolonged contact to skin
- Refer to SDS for additional warnings and instructions

For More Info:

SIPSEAL.COM | INFO@SIPSEAL.COM | (651) 300-0505

STATE OF CALIFORNIA
PROPOSITION 65
WARNING
This product may contain one or more substances or chemicals known to the state of California to cause cancer
California Health & Safety Code Sec 25249.5

INTERTEK SUSTAINABILITY
CERTIFIED
CLEAN AIR GOLD

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Vapor Tape EX Technical Data Sheet

Product Description

- Developed for SIPs and panelized construction as a durable, all weather vapor tight joint sealant
- Innovative hybrid synthetic rubber specially formulated to develop a strong bond with OSB and engineered wood products
- High initial tack for immediate bond
- Facing is thermally stable, smooth and conformable to irregular surfaces
- SIP-SEAL polyether products bond aggressively to the tape facer allowing for blind joint solutions and for use with SIP-SEAL Liquid Membrane.

Applications

- An air and vapor sealing tape for all types of construction
- Excellent adhesion to OSB, plywood, lumber, EPS foam, MgO, concrete, painted metal, most vinyl, PVC plastic and most common building materials (always test adhesion)

Technical Data

- 100% Solids - No VOC's
- Working temperature -40°F to 180°F (-40°C to 82°C)
- Application temperature 0°F to 120°F (-18°C to 49°C)
- Adhesion to OSB (tested to ASTM D-3330) 4.3#
- Adhesion to anodized aluminum (tested to ASTM D-3330) 4.9#
- Adhesion to house wrap (highly dependent on brand)
- Nail seal ability (Tested to AAMA 711-13) Pass
- Hydrophobic – will not absorb or attract moisture
- UV Stability – 6 months
- Resists common mold growth
- Vapor Tight
- Shelf life 24 months from date of manufacture

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Package/Color

- 4″ x 75′ Black 6/case (30310) – Back-slit O/C
- 6″ x 75′ Black 4/case (30311) – Back-slit O/C
- 12″ x 75′ Black 2/case (30313) – Back-slit 3″ from each edge
- 18″ x 75′ Black 1/case (30314) – Back-slit 3′ from each edge

Usage Instructions

- Apply to clean surfaces. For proper adhesion, bonding surface must be dry and free of debris, dust, dirt, frost or ice. It is advisable to use a stiff brush or scraper to remove loose particles of OSB from the bonding surface prior to application. Always check adhesion and chemical compatibility with any new substrate.
- Gradually peel back the release liner and begin applying short sections using hand pressure and working from the center out to avoid wrinkles. Overlap the tape at the tape joints for a tight seal.
- Use a J-roller, drywall blade or hard squeegee to work the tape into the surface and ensure that the edges are properly bonded.
- If tape does not adhere, additional site-specific measures should be taken to assure proper adhesion.

Limited Warranty

Products are free from manufacturing defects and when used according to the manufacturer's published specifications will meet the manufacturer's published performance standards for a period of ten years from the date of shipment from the factory. Sustant will, at its sole option, replace the defective product or refund the purchase price paid by the user with acceptable proof of purchase from an authorized seller. There are no other warranties, express or implied, including but not limited to any implied warranties of merchantability or fitness for a particular purpose. The user is responsible for determining the suitability of the product for the user's needs. Neither Sustant nor its manufacturers shall have a liability for any other damages, including but not limited to any indirect, special or consequential damages, even if they have been advised of the possibility of such.

For More Info:
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Made in USA

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Headlock Product Data Specifications

PRODUCT DESCRIPTION

The OMG HeadLok is a specialized, flat head fastener engineered for a wide range of panel applications including Structural Insulated Panels, Prefabricated Wall Panels, and Nailboard, and can also be used in wood, structural concrete*, purlins*, corrugated and structural steel substrates.

FEATURES & BENEFITS

• Three point and thread styles available for fast installation.
• Spade Point for use in in steel
(18 - 22 ga.), structural concrete and wood;
• Gimlet Point for use in dimensional lumber;
• Drill Point for use in steel purlins up to 3/16-in. thick. The 1/2-in. drill point allows the fastener to be drilled through the purlin before the threads engage.
• Aggressive thread for holding power and withdrawal strength.
• Chamfer under head for increased strength.
• Extra large flat head dramatically increases pull-through resistance and eliminates the need for additional washers.
• ACQ tested and approved.
• Install with SpiderDrive® bit system. Eliminates cam-out. Free drive bits
in every bucket.

COATING

OMG CR-10 corrosion resistant coating passes the corrosion requirements of FM Approval Standard 4470 and EAD030351-00-0402.

APPLICATION

Install the OMG HeadLok using a high torque, low RPM screw gun. Bring underside of the washer head flat to the surface. Do not overdrive. For steel substrates, proper point style must be determined depending on the steel gauge thickness. *F o r structural concrete, use the OMG HeadLok with spade points only. The fastener must penetrate structural concrete decks a minimum of 1-inch. Pre-drill using a 3/16-in. pilot hole at least 1/2-in. (13 mm) deeper than fastener embedment.
*For purlin attachment, use OMG HeadLok with drill point only.
*Prior to job-start, contact OMG to perform a pullout test to determine pullout values.

PHYSICAL DATA**

The data below is constant for each OMG HeadLok.

HEAD	POINT STYLES	SHANK
.625" (15.87 mm) Diameter 8 Lobe SpiderDrive® Bit	• Gimlet • Spade • Drill	.190" (4.82 mm) Diameter
		COATING
		CR-10

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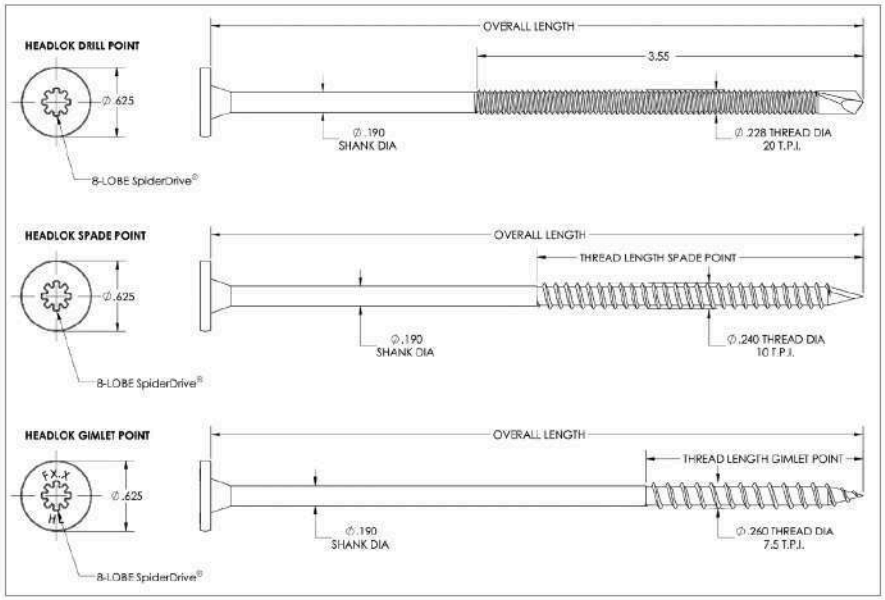
As with all building materials it is the responsibility of the installer to review its usage with a design professional to confirm safety, compatibility and acceptance with the roof cover manufacturer as well as local building codes.DRILL POINT



DRILL POINT			
CAT. NO.	LENGTH IN (MM)	THREAD LENGTH	PACK
FMHLDP006-250	6 (150)	3.5 (89)	250
FMHLDP008-250	8 (200)	3.5 (89)	250
FMHLDP010-250	10 (255)	3.5 (89)	250

SPADE POINT			
CAT. NO.	LENGTH IN (MM)	THREAD LENGTH	PACK
FMHLSP004-250	4 (100)	3 (76)	250
FMHLSP005-250	5 (125)	3 (76)	250
FMHLSP006-250	6 (150)	3 (76)	250
FMHLSP007-250	7 (175)	3 (76)	250
FMHLSP008-250	8 (200)	3 (76)	250
FMHLSP009-250	9 (230)	3 (76)	250
FMHLSP010-250	10 (255)	3 (76)	250

SPADE POINT			
CAT. NO.	LENGTH IN (MM)	THREAD LENGTH	PACK
FMHLGM278-500	2 7/8 (70)	2 (51)	500
FMHLGM334-250	3 3/4 (95)	2 (51)	250
FMHLGM412-250	4.5 (115)	2 (51)	250
FMHLGM005-250	5 (125)	2 (51)	250
FMHLGM006-250	6 (150)	2 (51)	250
FMHLGM007-250	7 (175)	2 (51)	250
FMHLGM008-250	8 (200)	2 (51)	250
FMHLGM009-250	9 (230)	2 (51)	250
FMHLGM010-250	10 (255)	2 (51)	250
FMHLGM011-250	11 (280)	2 (51)	250
FMHLGM012-250	12 (305)	2 (51)	250
FMHLGM013-250	13 (330)	2 (51)	250
FMHLGM014-250	14 (355)	2 (51)	250

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4.0 DESIGN AND INSTALLATION

4.1 Design: 4.1.1 Overall Structural System: The scope of this report is limited to the evaluation of the SIP component. Panel connections and other details related to incorporation of the product into the overall structural system of a building are beyond the scope of this report.
The seismic-force-resisting system of structures consisting of the panels as shear walls, in whole or in part, must be designed and detailed in accordance with IBC Sections 2305 and 2306.

4.1.2 Design Approval: Where required by the authority having jurisdiction, structures using Structural Insulated Panels shall be designed by a registered design professional. Construction documents, including engineering calculations and drawings providing floor plans, window details, door details and connector details, shall be submitted to the code official when application is made for a permit. The individual preparing such documents shall possess the necessary qualifications as required by the applicable code and the professional registration laws of the state where the construction is undertaken. Approved construction documents shall be available at all times on the jobsite during installation.

4.1.3 Design Loads: Design loads to be resisted by the product shall be as required under the applicable code. Loads on the panels shall not exceed the loads noted in this report. Where loading conditions result in superimposed stresses, the sum of the ratio of actual loads over allowable loads shall not exceed one. Calculations demonstrating that the loads applied are less than the allowable loads described in this report shall be submitted to the code official for approval.

4.1.4 Allowable Loads: Allowable axial, transverse and in-plane shear loads may be selected from Tables 3 through 13. For loading conditions not specifically addressed herein, structural members designed in accordance with accepted engineering practice shall be provided to meet applicable code requirements.

4.1.5 Concentrated Loads: Axial loads shall be applied to the product through continuous members such as structural insulated roof or floor panels or repetitive members such as joists, trusses or rafters spaced at regular intervals of 24 inches (610 mm) on center or less. Such members shall be fastened to a rim board or similar member to distribute the load to the
product. For other loading conditions, reinforcement shall be provided. This reinforcement shall be designed in accordance with accepted engineering practice.

4.1.6 Eccentric and Side Loads: Axial loads shall be applied concentrically to the top of the product. Loads shall not be applied eccentrically or through framing attached to one side of the panel (such as balloon framing) except where additional engineering documentation is provided.

4.1.7 Openings: Openings in panels are permitted when the header depth is at least 12 inches (305 mm), and the interior of the opening is reinforced with minimum 0.42 SG lumber graded #2 around the perimeter, secured in place with not less than 0.131-inch x 21/2-inch (2.9 mm x 63.5 mm) nails, spaced 6 inches (152 mm) on center. The panels are not used to resist in-plane shear loads. SIP splines are not permitted within 6 inches of the end of the header and are not permitted within the header. Allowable loads for maximum header spans of 36 inches may be selected from Tables 8 and 10. Allowable loads for maximum header spans of 72 inches (1829 mm) may be selected from Tables 9 and 11. Openings in panels beyond the scope of this report shall be reinforced with wood or steel designed in accordance with accepted engineering practice to resist all loads applied to the opening as required by the adopted code. Details for door and window openings shall be provided to clarify the manner of supporting axial, transverse and/or in-plane shear loads at openings. Such details shall be subject to approval by the local authority having jurisdiction.
4.1.8 In-Plane Shear Design: Shear walls utilizing block or surface splines shall be sized to resist all code required wind and seismic loads without exceeding the allowable loads provided herein. Shear wall chords, hold-downs and connections to transfer shear forces between the wall and surrounding structure shall be designed in accordance with accepted engineering practice. Allowable strengths for SIP shear walls with structural splines along each panel edge shall be designed in accordance with accepted engineering practice and are subject to the limitations for wood sheathed shear walls.

4.1.8.1 Seismic Design Categories A, B, and C: Use of the shear wall configurations in Table 12 is limited to structures in Seismic Design Categories A, B and C. Where SIPs are used to resist seismic forces the following factors shall be used for design: Response Modification Coefficient, R = 2.0; System Overstrength Factor, $\Omega 0$ = 2.5; Deflection Amplification Factor, Cd = 2.0. The maximum panel height-to-width ratio shall be 2:1.

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4.1.8.2 Seismic Design Categories D, E and F: SIPs may be used in seismic-force-resisting systems where the panels are non-load bearing. SIPs constructed in accordance with Table 13 used in Seismic Design Categories C, D, E and F use the following seismic design parameters; R=6 ½, Ω0=3 and Cd=4. SIPs used in Seismic Design Categories C through F must be a minimum of 8 ft tall and be continuous (no spline connections) when used in aspect ratios of 1:1 and greater. Walls may be constructed using surface or block splines a minimum of 4 ft (1219 mm) on center when the aspect ratio of the wall is 0.4:1 or less.

4.1.9 Horizontal Diaphragms: Horizontal diaphragms shall be sized to resist all code required wind and seismic loads without exceeding the allowable loads provided herein. Diaphragm chords and connections to transfer shear forces between the diaphragm and surrounding structure shall be designed in accordance with accepted engineering practice. The maximum diaphragm length-to-width ratio shall not exceed 3:1.

4.1.10 Combined Loads: Panels subjected to any combination of transverse, axial or in-plane shear loads shall be analyzed utilizing a straight-line interaction.

4.1.11 Panel Reinforcements: Allowable transverse loads for panels reinforced with I-joists meeting the minimum properties shown in Table 2 are presented in Table 6. Panels reinforced with I-joists have not been evaluated for use in wall applications. Panels reinforced with I-joist splines may be designed in accordance with accepted engineering practice.

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4.2 Installation:

4.2.1 General: Structural Insulated Panels shall be fabricated, identified and erected in accordance with this report, the approved construction documents and the applicable codes. In the event of a conflict between the manufacturer's published installation instructions and this report, this report shall govern. Approved construction documents shall be available at all times on the jobsite during installation.

4.2.2 Splines: Structural Insulated Panels are interconnected at the panel edges through the use of a spline. The spline type may be of any configuration listed in Section 3.2.5 as required by the specific design. The spline shall be secured in place with not less than 0.131-inch x 21/2-inch (2.9 mm x 63.5 mm) nails, spaced 6 inches on center on both sides of the panel, or an approved equivalent fastener. All joints shall be sealed in accordance with the SIP manufacturer's installation instructions. Alternate spline connections may be required for panels subjected to in-plane shear forces. Such panels shall be interconnected exactly as required in Tables 12 and 13 or as directed by the designer.

4.2.3 Plates: The top and bottom plates of the panels shall be dimensional or engineered lumber sized to match the core thickness of the panel. The plates shall be secured using not less than 0.131-inch x 21/2-inch (2.9 mm x 63.5 mm) nails, spaced 6 inches on center on both sides of the panel, or an approved equivalent fastener. A second top plate of 11/8-inch (29 mm) minimum thickness dimensional or engineered lumber with a specific gravity of 0.42 that is cut to the full thickness of the panel shall be secured to the first top plate using 0.131-inch x 3-inch (2.9 mm x 76 mm) nails or an approved equivalent fastener.

4.2.4 Cutting and Notching: No field cutting or routing of the panels shall be permitted except as shown on approved construction documents.

4.2.5 Protection from Decay: SIPs that rest on exterior foundation walls shall not be located within 8 inches of exposed earth. SIPs supported by concrete or masonry that is in direct contact with earth shall be protected from the concrete or masonry by a moisture barrier.

4.2.6 Protection from Termites: In areas subject to damage from termites, SIPs shall be protected from termites using an approved method. Panels shall not be installed below grade or in contact with earth.

4.2.7 Heat-Producing Fixtures: Heat-producing fixtures shall not be installed in the panels unless protected by a method approved by the code official or documented in test reports. This limitation shall not be interpreted to prohibit heat-producing elements with suitable protection.

4.2.8 Plumbing Installation Restrictions: Plumbing and waste lines may extend at right angles through the wall panels but are not permitted vertically within the core. Lines shall not interrupt splines or panel plates unless approved by a registered design professional.

4.2.9 Voids and Holes: 4.2.9.1 Voids in Core: In lieu of openings designed in accordance with Section 4.1.7, the following voids are permitted. Voids may be provided in the panel core during fabrication at predetermined locations only. Voids parallel to the panel span shall be limited to a single 1-inch-maximum-diameter (25.4 mm) hole. Such voids shall be spaced a minimum of 4 feet (1219 mm) on center measured perpendicular to the panel span. Two 1/2-inch-diameter (12.7 mm) holes may be substituted for the single 1-inch hole provided they are maintained parallel and within 2 inches of each other. Voids perpendicular to the panel span shall be limited to a single 1-inch-maximum-diameter (25.4 mm) hole placed not closer than 16 inches (406 mm) from the support. Additional voids in the same direction shall be spaced not less than 28 inches (711 mm) on center.

4.2.9.2 Holes in Panels: Holes may be placed in panels during fabrication at predetermined locations only. Holes shall be limited to 4 inches by 4 inches (102 mm by 102 mm) square. The minimum distance between holes shall not be less than 4 feet (1219 mm) on center measured perpendicular to the panel span and 24 inches (610 mm) on center measured parallel to the panel span. Not more than three holes shall be permitted in a single line parallel to the panel span. The holes may intersect voids permitted elsewhere in this report.

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4.2.10 Panel Cladding: 4.2.10.1 Roof Covering: The roof covering, underlayment and flashing shall comply with the applicable codes. All roofing materials must be installed in accordance with the manufacturer's installation instructions. The use of roof coverings requiring the application of heat during installation shall be reviewed and approved by a registered design professional.

4.2.10.2 Exterior Wall Covering: Panels shall be covered on the exterior by a water-resistive barrier as required by the applicable code. The water-resistive barrier shall be attached with flashing in such a manner as to provide a continuous water-resistive barrier behind the exterior wall veneer. The exterior facing of the SIP wall shall be covered with weather protection as required by the adopted building code or other approved materials.

4.2.10.3 Thermal Barrier at Wall, Roof and Floor: The foam plastic core of the SIPs must be separated from the interior of the building by installing an approved thermal barrier on the interior face of the panels consisting of ½-inch (12.7 mm) gypsum wallboard or equivalent thermal barrier complying with and installed in accordance with IBC Section 2603.4 or IRC Section R316.4 as applicable.

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4.3 Fire-Resistance Rated Assemblies: 4.3.1 One-hour Limited Load-Bearing Wall Assembly: SIPs with thicknesses of 4 5/8 . 6 ½ , or 8 ¼ inches (114, 165, or 210 mm) are used to construct a one-hour fire-resistance rated wall assembly. The SIP core is recessed 1-1/2 inches (38 mm) from the bottom SIP edge and 1-1/2 inches (38 mm) from the top SIP edge. The recesses receive nominally 2-by spruce-pine-fir No. 2 or better lumber bottom and top plates with a depth to match the core thickness. The plates must be connected to the SIPs by fastening through the SIP OSB facing with 8d box nails spaced 6 inches (152 mm) on center, on each side of the SIP.

The SIP core is recessed on the vertical sides to receive surface or block splines in accordance with Section 3.2.5 of this report. The splines must be connected to the SIPs by fastening through the SIP OSB facing with 1 5/8-inch long (41.28 mm), Type W, Self-piercing tapping screws (ASTM C1002) spaced 6 inches (152 mm) on center.
The SIPs must be covered with two layers of 5/8-inch thick (15.9 mm) Type X gypsum wallboard, complying with ASTM C1396, on each side. Where the panels are exposed to the exterior, the exterior layers of gypsum boards must be 5/8-inch thick (15.9 mm) Type X gypsum sheathing complying with ASTM C1396. The vertical joints of the first layer of gypsum board must be offset a minimum of 16 inches (406 mm) from the spline joint. The first layer of gypsum board must be fastened to the panel facing with 1 5/8-inch long (41.28 mm), Type W, self-piercing tapping screws complying with ASTM C1002, spaced 24 inches (610 mm) on center vertically and 16 inches (406 mm) on center horizontally. The second layer of gypsum board must be installed with 2-inch long (50.8 mm), Type W, self-piercing tapping screws complying with ASTM C1002, spaced 12 inches (305 mm) on center vertically in rows offset 12 inches (305 mm) from screws securing the first layer of gypsum board, and 16 inches (406 mm) on center horizontally, in rows offset 8 inches (203 mm) from screws securing the first layer of gypsum board. The vertical joints in the second layer of gypsum board must be offset a minimum of 16 inches (406 mm) from vertical joints of the first layer of gypsum board.
Exposed gypsum board joints must be covered with joint tape and joint compound. Exposed screw heads must be covered with joint compound in accordance with ASTM C840.
This fire-resistance-rated wall assembly is limited to the heights shown in Table 7. The maximum superimposed allowable axial compression load shall be no greater than the lesser of 1800 plf (26 kN/m) or 43% of the loads in Table 7.

4.3.2 One-hour Limited Load-Bearing Wall Assembly: SIPs in accordance with this report, with thickness of 6 ½ inch (165 mm) and up to 10 feet (3048 mm) tall are used to construct a one-hour fire-resistance rated wall assembly. The SIP core is recessed 1 ½ inches (38 mm) from the bottom and side SIP edge. The SIP core is recessed 3 inches (76 mm) from the top SIP edge. The recesses receive 2x6 #2 Hem-Fir lumber; single bottom plate, double top plate, and single lumber end plates and double lumber splines. The double lumber splines are attached to each other using 0.148 x 3.25 inch (4x83 mm) nails with 0.35 inch (9 mm) head diameters, 24 inches (609 mm) on center, staggered.

The first top plate is attached to the studs using (2) 0.148 x 3.25 inch (3.8x82.6 mm) nails with 0.35 inch (8.9 mm) head diameters. The second top plate is attached to the first top plate using 0.148 x 3.25 inch (3.8x82.6 mm) nails with 0.35 inch (8.9 mm) head diameters, 16 inches (406 mm) on center.

OSB facing is fastened to perimeter framing using 0.122 x 2.08 inch (3x53 mm) with 0.25 inch (6 mm) head diameter, 6 inch (152 mm) on center.
The SIPs must be covered with a single layer of 5/8 Type C gypsum on oriented vertically on both sides of the wall, secured with PC cupped head drywall nails 1 5/8 inch (41 mm) long, 8 inch (203 mm) on center along the perimeter and 12 inch (304 mm) on center through the field. The fastener heads and joints must be treated with joint compound and paper tape.
The wall assembly is limited to a maximum restricted superimposed load of 56% of the allowable capacity shown in Table 7.

4.3.3 One-hour Roof-Ceiling Assembly: SIP with thicknesses from 4 5/8 inches to 12 ¼ inches (114 mm to 286 mm) thick. Support beams must be a minimum of 4 ½ inches wide by 9 ½ inches deep (114 mm by 241 mm) and must be spaced in accordance with the IBC or IRC as applicable. The roof covering material must comply with the IBC. The roof construction must comply as a Class A, B, or C roof assembly. SIPs must be connected with double nominal 2-inch lumber splines installed in the recessed core and connected to the SIP by fastening through the OSB facing with 8d common nails spaced 6 inches (31.7 mm) on center. Double lumber splines are attached to each other using 12d nails applied in 2 rows, 10 inches on center. Each exposed SIP edge must be covered with nominally 2-inch wood blocking installed in the recessed core and connected to the SIP by fastening through the OSB facing with 8d common nails spaced 6 inches (152 mm) on center.

Two layers of minimum 5/8 inch (15.9 mm) thick gypsum board complying with ASTM C1396 must be installed on the underside of the SIPs and wood beams. The gypsum board's long dimension must be installed perpendicular to the wood beams. The first layer must be connected using 1 ¼ inch long (31.7 mm), Type S, bugle head steel screws complying with ASTM C1002, spaced 8 inches (203

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mm) on center along the joints and in rows spaced 16 inches (406 mm) on center in the field. The joints of the first layer of gypsum must be staggered from the joints of the SIPs. The second layer of gypsum board must be fastened using a 2 inch long (51 mm), bugle head. Type W, self-piercing steel screws complying with ASTM C1002, spaced 8 inches (204 mm) on center along the board edges and in rows 12 inches (305 mm) on center in the field. The joints of the gypsum board second layer must be staggered from the joints of the gypsum board first layer. The exposed gypsum board joints must be covered with paper tape and joint compound. Screw heads must be covered with joint compound in accordance with ASTM C840.
The roof-ceiling assembly is limited to a maximum transverse load of the lesser of 43 psf (2.06 kPa) or the loads in Table 3.

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4.4 Special Inspection: Where SIP shear walls are installed in buildings in Seismic Design Categories C, D, E and F; Seismic Design Categories C, D0, D1, D2 and E for townhouses under the IRC; or Seismic Design Categories D0, D1, D2 and E for detached one- and two-family dwellings under the IRC, periodic inspections of the fastening and anchoring of the shear wall assembly withing the seismic-force-resisting system must be provided. Inspection must include connections of the assemblies to drag struts and hold-downs, in accordance with 2018 and 2015 IBC Section 1705.11.1 or 1705.12.2, 2012 IBC Section 1705.10.1 or 1705.11.2, 2009 IBC Section 1706.2 or 1707.3, or 2006 IBC Section 1707.3, as applicable, unless these are exempted by IBC Section 1704.1.

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5.0 CONDITIONS OF USE:

The Structural Insulated Panels described in this report comply with, or are a suitable alternative to what is specified in, those codes listed in Section 1.0 of this report, subject to the following conditions:

5.1 The SIPs are fabricated, identified, and erected in accordance with this report, the manufacturer's published installation instructions and the approved construction documents. If there is a conflict between this report and the manufacturer's instructions, the more restrictive governs.

5.2 This report applies only to the panel thicknesses specifically listed herein.

5.3 In-use panel heights/spans shall not exceed the values listed herein. Extrapolation beyond the values listed herein is not permitted.

5.4 Design loads to be resisted by the SIPs must be determined in accordance with the IBC or IRC, as applicable, and must not exceed the allowable loads noted in this report.

5.5 The panels are manufactured at the production facilities listed in Section 7.3 of this evaluation report under a quality-control program with inspections by ICC-ES.

5.6 Where the SIP panels are used as a shear wall assembly for buildings located in Seismic Design Category D, E, or F under the IBC, for townhouses in Seismic Design Categories C, D0, D1, D2 and E under the IRC; or for detached one-and two-family dwellings in Seismic Design Categories D0, D1, D2 and E under the IRC, and are either of different widths intended to be combined in the same wall line or are intended to be combined in the same wall line with other shear-resisting elements, applied lateral loads shall be proportioned based on relative stiffness.

5.7 Seismic-force-resisting systems consisting of SIP shear walls in whole or in part shall be designed and detailed in accordance with Sections 2305 and 2306 of the IBC by the registered design professional.

5.8 Calculations and details must be submitted to the code official showing how the lateral loads are transferred from the roof or floor diaphragm into the shear wall and from the shear wall to the foundation. These calculations and details must be signed and sealed by a registered design professional, when required by the statures of the jurisdiction in which the project is to be constructed.

5.9 When the SIP shear walls are used in buildings that are more than one story tall, calculations and details must be submitted to the code official showing the load path for the transfer of lateral and overturning forces from the upper-story shear walls to the foundation. These calculations and details must be signed and sealed by a registered design professional, when required by the statutes of the jurisdiction in which the project is to be constructed.

5.10 Shear walls constructed of SIPs, used in buildings in Seismic Design Categories C through F, must be subject to special inspection in accordance with Section 4.3.

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6.0 EVIDENCE SUBMITTED

6.1 Data in accordance with ICC_ES AC04 Acceptance Criteria for Sandwich Panels Approved June 2019 (Editorially revised December 2020).

6.2 Reports of diaphragm tests of panels, conducted in accordance with ASTM E455.

6.3 Reports and analysis of cyclic shear wall testing in accordance with Appendix A of the ICC-ES Acceptance Criteria for Sandwich Panels (AC04) dated June 2019 (editorially revised December 2020).

6.4 Reports of tests conducted in accordance with ASTM E119.

6.5 Reports of tests conducted in accordance with NFPA 286.

7.0 IDENTIFICATION

7.1 Structural Insulation Panels are identified with the following information: **7.1.1** The ICC-ES Evaluation Report number (ESR-4689).

7.1.2 Project or batch number

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TABLE 1—OSB FACING MINIMUM PROPERTIES

Thickness (in.)	Flatwise Stiffness (lbf-in.2/ft)		Flatwise Strength (lbf-in./ft)		Tension (lbf/ft)		Density (pcf)
	Along	Across	Along	Across	Along	Across	
7/16	54,700	27,100	950	870	6,800	6,500	35

For SI: 1 inch = 25.4 mm; 1 foot = 304.8 mm; 1 lbf = 4.448 N; 1 pcf = 0.006366 N/m3; 1 lbf-in/ft = 370.833 N-mm/m; 1 lbf/ft = 14.59 N/m; 1 lbf-in.2/ft = 9419.167 N-mm/m

TABLE 2—MINIMUM I-JOIST PROPERTIES FOR USE AS REINFORCEMENTS[1]

Depth	Bending Stiffness EI	Moment Capacity M	Shear Capacity V	Coefficient of Shear Deflection K
(In.)	(lbf-in.2) x 106	(lbf-ft)	(lbf)	(lbf) x 106
9.35	185	2715	1155	4.81
11.25	296	3410	1405	5.85
14	482	4270	1710	7.28

For SI: 1 inch = 25.4 mm; 1 foot = 304.8 mm; 1 lbf = 4.448 N; 1lbf-in.2 = 2870.962 N-mm

[1] Properties are based on certification in accordance with ASTM D5055 or equivalent.

TABLE 3—ALLOWABLE ROOF UNIFORM TRANSVERSE LOADS, BLOCKED BEARING, SHORT DURATION (PSF) [1, 4]

Panel Length (ft)	PANEL THICKNESS (inch)								
	4 5/8			6 1/2			8 1/4		
	Deflection Limit[2]			Deflection Limit[2]			Deflection Limit[2]		
	L/180	L/240	L/360	L/180	L/240	L/360	L/180	L/240	L/360
8 WAB[3]	50	40	27	73	64	43	80	80	58
8	68	51	34	82	82	56	90	90	78
10	45	33	22	63	57	38	68	68	54
12	30	23	15	51	40	27	55	55	39
14	21	16	--	39	29	19	46	43	29
16	--	--	--	29	22	14	40	33	22
18	--	--	--	22	16	--	34	25	17
20	--	--	--	--	--	--	26	20	13
22	--	--	--	--	--	--	21	15	--
24	--	--	--	--	--	--	17	12	--

For SI: 1 inch = 25.4 mm; 1 foot = 304.8 mm; 1 psf = 47.88 Pa.

See Table 4 for notes.

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TABLE 4—ALLOWABLE ROOF UNIFORM TRANSVERSE LOADS, BLOCKED BEARING, SHORT DURATION (PSF) [1, 4]

Panel Length (ft)	PANEL THICKNESS (inch)								
	10 1/4			12 1/4			15		
	Deflection Limit[2]			Deflection Limit[2]			Deflection Limit[2]		
	L/180	L/240	L/360	L/180	L/240	L/360	L/180	L/240	L/360
8 WAB[3]	88	88	75	93	96	96	108	108	108
8	98	98	98	107	107	107	121	121	121
10	73	73	73	79	79	79	87	87	87
12	59	59	54	63	63	63	68	68	68
14	49	49	41	52	52	52	56	56	56
16	42	42	31	44	44	41	47	47	47
18	37	36	24	39	39	32	41	41	41
20	32	29	19	34	34	26	36	36	36
22	29	23	15	31	31	21	33	33	29
24	25	19	12	28	26	17	29	29	24

*For **SI:** 1 inch = 25.4 mm; 1 foot = 304.8 mm; 1 psf = 47.88 Pa.*

1 Table values assume a simply supported panel with 1 1/2 in. of continuous bearing on facing at supports (C_p = 1.0) with solid wood plates at bearing locations. Values do not include the dead weight of the panel.

2 Deflection limit shall be selected by building designer based on the serviceability requirements of the structure and the requirements of adopted building code. Values are based on loads of short duration only and do not consider the effects of creep.

3 Tabulated values are based on the strong-axis of the facing material oriented parallel to the direction of panel bending. WAB indicates weak-axis bending of the facing material; the strong-axis of the facing material is oriented perpendicular to the direction of panel bending.

4 Permanent loads, such as dead load, shall not exceed 0.50 times the tabulated load.

TABLE 5—ALLOWABLE WALL UNIFORM TRANSVERSE LOADS (PSF) [1, 4]

Panel Length (ft)	PANEL THICKNESS (inch)								
	4 5/8			6 1/2			8 1/4		
	Deflection Limit[2]			Deflection Limit[2]			Deflection Limit[2]		
	L/180	L/240	L/360	L/180	L/240	L/360	L/180	L/240	L/360
8 WAB[3]	62	50	35	24	24	24	49	49	49
8	66	56	39	63	63	56	62	62	62
10	52	43	30	55	55	48	57	56	55
12	37	30	21	48	46	40	51	50	49
14	23	17	12	40	38	33	46	44	42
16	--	--	--	32	30	25	40	38	36
18	--	--	--	25	21	17	35	32	29
20	--	--	--	17	13	9	29	26	22
24	--	--	--	--	--	--	18	14	9

*For **SI:** 1 inch = 25.4 mm; 1 foot = 304.8 mm; 1 psf = 47.88 Pa.*

1 Table values represent wall panel capacities (4 5/8-in., 6 1/2-in. and 8 1/4-in. thickness panels only) utilizing a zero bearing configuration (Figure 2).

2 Deflection limit shall be selected by building designer based on the serviceability requirements of the structure and the requirements of adopted building code. Values are based on loads of short duration only and do not consider the effects of creep.

3 Tabulated values are based on the strong-axis of the facing material oriented parallel to the direction of panel bending. WAB indicates weak-axis bending of the facing material; the strong-axis of the facing material is oriented perpendicular to the direction of panel bending.

4 Permanent loads, such as dead load, shall not exceed 0.50 times the tabulated load.

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TABLE 6—ALLOWABLE UNIFORM TRANSVERSE LOADS WITH I-JOIST REINFORCEMENTS (PSF) [1, 3, 4]

Panel Length (ft)	PANEL THICKNESS (inch)								
	10 1/4-in. SIP thickness			12 1/4-in. SIP thickness			15-in. SIP thickness		
	Deflection Limit[2]			Deflection Limit[2]			Deflection Limit[2]		
	L/180	L/240	L/360	L/180	L/240	L/360	L/180	L/240	L/360
8	115	115	115	124	124	124	123	123	123
10	92	92	92	99	99	99	98	98	98
12	76	76	76	82	82	82	82	82	82
14	65	65	65	71	71	71	70	70	70
16	57	57	57	62	62	62	61	61	61
18	51	51	44	55	55	55	54	54	54
20	46	46	33	49	49	48	48	48	48
22	41	38	25	45	45	37	44	44	44
24	36	30	20	41	41	29	41	41	41

*For **SI:** 1 inch = 25.4 mm; 1 foot = 304.8 mm; 1 psf = 47.88 Pa.*

1 Values assume a simply supported panel with 1 1/2 in. of continuous bearing on facing at supports. Values do not include the dead weight of the panel.

2 Deflection limit shall be selected by building designer based on the serviceability requirements of the structure and the requirements of adopted building code. Values are based on loads of short duration only and do not consider the effects of creep.

3 Tabulated values are based on the strong-axis of the facing material oriented parallel to the direction of panel bending.

4 Permanent loads, such as dead load, shall not exceed 0.50 times the tabulated load.

TABLE 7—ALLOWABLE AXIAL LOADS (PLF) [1,2,3,4]

Panel Length (ft)	PANEL THICKNESS (inch)		
	L/180	L/240	L/360
8 WAB[5]	2320	2470	2530
8	3630	4070	4240
10	3260	3890	4130
12	2810	3660	4000
14	--	3390	3830
16	--	3090	3640
18	--	2790	3430
20	--	--	3190

*For **SI:** 1 inch = 25.4 mm; 1 foot = 304.8 mm; 1 PLF = 14.59 N/m.*

1 Permanent loads, such as dead load, shall not exceed 0.50 times the tabulated load.

2 All values are for normal duration and may not be increased for other durations.

3 Axial loads shall be applied concentrically to the top of the panel through repetitive members spaced not more than 24-in. on center. Such members shall be fastened to a rim board or similar member to distribute along the top of the SIP.

4 The ends of both facings must bear on the supporting foundation or structure to achieve the tabulated axial loads.

5 Tabulated values are based on the strong-axis of the facing material oriented parallel to the direction of panel bending. WAB indicates weak-axis bending of the facing material; the strong-axis of the facing material is oriented perpendicular to the direction of panel bending.

6 For fire-resistance-rated wall assemblies, axial load limitations in Section 4.3 must be observed.

7 For combined loading, the requirements in Section 4.1.10 must be applied.

8 The maximum allowable axial load is limited to 71 percent of the reported allowable axial load when used in the shear walls in Table 13.

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TABLE 8—ALLOWABLE UNIFORM TRANSVERSE LOADS FOR SIPS WITH OPENINGS, 36-IN. MAXIMUM SPAN (PSF) [1,4,5,6]

Panel Length (ft)	PANEL THICKNESS (inch)								
	4 5/8			6 1/2			8 1/4		
	Deflection Limit[2]			Deflection Limit[2]			Deflection Limit[2]		
	L/180	L/240	L/360	L/180	L/240	L/360	L/180	L/240	L/360
8 WAB[3]	23	17	11	42	31	21	62	47	31
8	31	23	15	57	43	28	75	65	43
10	17	13	8	33	25	16	48	39	26
12	10	8	5	21	16	10	33	25	16
14	7	5	--	14	10	7	22	16	11
16	--	--	--	9	7	--	15	11	7
18	--	--	--	7	5	--	11	8	5
20	--	--	--	--	--	--	8	6	--

For **SI:** 1 inch = 25.4 mm; 1 foot = 304.8 mm; 1 psf = 47.88 Pa.
See Table 9 for notes.

TABLE 9—ALLOWABLE UNIFORM TRANSVERSE LOADS FOR SIPS WITH OPENINGS, 72-INCH MAXIMUM SPAN (PSF) [1,4,5,6]

Panel Length (ft)	PANEL THICKNESS (inch)								
	4 5/8			6 1/2			8 1/4		
	Deflection Limit[2]			Deflection Limit[2]			Deflection Limit[2]		
	L/180	L/240	L/360	L/180	L/240	L/360	L/180	L/240	L/360
8 WAB[3]	16	12	8	29	23	15	39	36	24
8	23	17	11	37	33	22	49	49	34
10	12	9	6	24	19	12	31	29	19
12	7	5	--	15	11	7	21	18	12
14	5	--	--	10	7	5	16	12	8
16	--	--	--	7	5	--	11	8	5
18	--	--	--	5	--	--	8	6	--
20	--	--	--	--	--	--	6	--	--

For SI: 1 inch = 25.4 mm; 1 foot = 304.8 mm; 1 psf = 47.88 Pa.

1 Table values represent wall panel capacities utilizing a zero bearing configuration (Figure 2). Construction shall be as described in Section 4.1.7 of this report.
2 Deflection limit shall be selected by building designer based on the serviceability requirements of the structure and the requirements of adopted building code. Values are based on loads of short duration only and do not consider the effects of creep.
3 Tabulated values are based on the strong-axis of the facing material oriented parallel to the direction of panel bending. WAB indicates weak-axis bending of the facing material; the strong-axis of the facing material is oriented perpendicular to the direction of panel bending.
4 Permanent loads, such as dead load, shall not exceed 0.50 times the tabulated load.

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5 Tabulated values assume header depths ranging from 12-in. to 36-in.
6 SIP splines are not permitted within 6-in. of the end of the header and are not permitted within the header.

TABLE 10—ALLOWABLE AXIAL LOADS FOR SIPS WITH OPENINGS, 36-IN. MAXIMUM SPAN (PLF) 1,2,3,4,6,7

Lateral Brace Spacing (ft)	PANEL THICKNESS (inch)		
	4 5/8	6 1/2	8 1/4
8 WAB5	770	820	840
8	1210	1355	1410
10	1085	1295	1375
12	935	1220	1330
14	--	1130	1275
16	--	1030	1210
18	--	930	1140
20	--	--	1060

For **SI:** 1 inch = 25.4 mm; 1 foot = 304.8 mm; ; 1 plf = 14.59 N/m.
See Table 11 for notes.

TABLE 11—ALLOWABLE AXIAL LOADS FOR SIPS WITH OPENINGS, 72-IN. MAXIMUM SPAN (PLF) 1,2,3,4,6,7

Lateral Brace Spacing (ft)	PANEL THICKNESS (inch)		
	4 5/8	6 1/2	8 1/4
8 WAB5	460	490	505
8	725	810	845
10	650	775	825
12	560	730	800
14	--	675	765
16	--	615	725
18	--	555	685
20	--	--	635

For SI: 1 inch = 25.4 mm; 1 foot = 304.8 mm; 1 plf = 14.59 N/m.
1 Permanent loads, such as dead load, shall not exceed 0.50 times the tabulated load.
2 All values are for normal duration and may not be increased for other durations.
3 Axial loads shall be applied concentrically to the top of the panel through repetitive members spaced not more than 24-in. on center. Such members shall be fastened to a rim board or similar member to distribute along the top of the SIP.
4 The ends of both facings must bear on the supporting foundation or structure to achieve the tabulated axial loads.
5 Tabulated values are based on the strong-axis of the facing material oriented parallel to the direction of panel bending. WAB indicates weak-axis bending of the facing material; the strong-axis of the facing material is oriented perpendicular to the direction of panel bending.
6 Tabulated values assume header depths ranging from 12-in. to 36-in.
7 SIP splines are not permitted within 6-in. of the end of the header and are not permitted within the header.

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TABLE 12—ALLOWABLE IN-PLANE SHEAR STRENGTH (POUNDS PER FOOT) FOR SIP SHEAR WALLS (WIND AND SEISMIC LOADS IN SEISMIC DESIGN CATEGORIES A, B AND C) [1, 2]

Spline Type[3]	Minimum Nominal SIP Thickness (in.)	Minimum Facing Connections[2,4]			Shear Strength(plf)
		Chord[2]	Plate[2]	Spline[3]	
Block or Surface Spline	4 5/8	0.131-in. x 2 1/2-in. nails, 6-in. on center	0.131-in. x 2 1/2-in. nails, 6-in. on center	0.131-in. x 2 1/2-in. nails, 6-in. on center	380
	8 1/4	0.131-in. x 2 1/2-in. nails, 6-in. on center	0.131-in. x 2 1/2-in. nails, 6-in. on center	0.131-in. x 2 1/2-in. nails, 6-in. on center	400

For **SI:** 1 inch = 25.4 mm; 1 foot = 304.8 mm; 1 psf = 47.88 Pa.; 1 plf = 14.59 N/m.
1 Maximum shear wall dimensions ratio shall not exceed 2:1 (height: width) for resisting wind or seismic loads.
2 Chords, hold downs and connections to other structural elements must be designed by a registered design professional in accordance with accepted engineering practice.
3 Spline type at interior panel-to-panel joints only. Solid chord members are required at each end of each shear wall segment.
4 Required connections must be made on each side of the panel. Dimensional or engineered lumber shall have an equivalent specific gravity of 0.42 or greater.

TABLE 13 – SHEAR WALLS FOR USE IN SEISMIC DESIGN CATEGORIES A, B, C, D, E AND F[3, 8, 9]

Spline Type	Minimum Nominal Thickness (in)	Aspect Ratio (H:W)	Minimum Facing Connections			Allowable Loads (plf)
			Sheathing to Chords	Sheathing to Top and Bottom	Sheathing to Splines	
[1,4,6,10] 23/32 inch OSB Surface or Block Spline	6 ½	1:1	0.113 x 2.375 inch nails, 2 inch on center, (2) rows ¾ and 2-1/4 inch edge distance	0.113 x 2.375 inch nails, 2 inch on center, (2) rows ¾ and 2-1/4 inch edge distance top plate and 3/8 inch and ¾ inch edge distance for bottom plate	0.113 x 2.375 inch nails, 2 inch on center, (2) rows 3/8 inch and ¾ inch edge distance	911
[2,5,7,11] 23/32 inch OSB Surface or Block Spline	6 ½	1:1, 2:1, 3.5:1	0.131 x 2.5 inch nails, 4 inch on center, ¾ inch edge distance.	0.131 x 2.5 inch nails, 4 inch on center, ¾ inch edge distance.	0.131 x 2.5 inch nails, 4 inch on center, ¾ inch edge distance.	803

1 Maximum aspect ratio of 1:1.
2 Maximum aspect ratio of 3.5:1. When the aspect ratio exceeds 2:1 the panel may not contain a spline joint.
3 Hold downs and connections to other structural elements must be designed by a registered design professional in accordance with accepted engineering practice.
4 Top plate consists of (2) 2x6 #2 Douglas Fir let-in framing stitched together using 0.148 x 3 inch nails 5 inch on center , two rows. Cap plate consist of (1) 1.125 x 6.375 inch OSB rim board attached to top plate using 0.131 x 3 inch nails, 6 inch on center, two rows. Bottom plate consists of (1) 2x6 #2 Douglas Fir let-in framing. Sill plate consists of (1) 2x8 #2 Douglas Fir attached to the bottom plate using 0.131 x 3 inch nails, 6 inch on center, two rows.
5 Top and bottom plates consist of (1) 2x6 #2 Douglas Fir let-in framing with an additional cap or sill plate. Sill plate consists of (1) 2x8 #2 Douglas Fir attached to the bottom plate using 0.131 x 3 inch nails, 6 inch on center, two rows. Cap plate consist of (1) 1.125 x 6.375 inch OSB rim board attached to top plate using 0.131 x 3 inch nails, 6 inch on center, two rows.
6 End post/Chords consist of 4x6 #1 Douglas Fir, attached to bottom plate using (3) 0.162 x 3.5 inch nails, toenailed through each chord into the bottom plate and fastened to the top plate using (6) 0.162 x 3.5 inch nails through the top plate into each chord.
7 End post/Chords consist of (2) 2x6 #2 Douglas Fir stitched together using (2) rows of 0131 x 3 inch nails 6 inch on center. Chords shall be attached to bottom plate using (3) 0.162 x 3.5 inch nails, toenailed through each chord into the bottom plate and fastened to the top plate using (3) 0.162 x 3.5 inch nails through the top plate into each chord.
8 The installation configuration is recognized for use as both load-bearing and non-load-bearing shearwalls in Seismic Design Categories A, B, C, D, E, and F with the seismic design coefficients of R=6.5; System Overstrength Factor, Ω_0=3.0; Deflection Amplification Factor, C_d=4.0.
9 When used as load-bearing SIPs, the allowable axial load must be determined in accordance with Table 7 of this report.
10 Figure 4 contains an illustration of the construction.
11 Figure 5 contains an illustration of the construction.

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TABLE 14—ALLOWABLE IN-PLANE SHEAR STRENGTH FOR HORIZONTAL DIAPHRAGMS SUBJECTED TO WIND OR SEISMIC LOADING

Minimum Nominal SIP Thickness (in.)	Minimum Connections			Shear Strength (plf)	Max. Aspect Ratio
	Surface Spline[1] (Figure 3b)	Boundary Support Element (Figure 3c)	Interior Support Spline[2,3] (Figure 3a)		
8-1/4	0.131-in. x 2 1/2-in. nails, 6-in. on center 7/16-in. x 3-in. OSB Surface Spline	10-in. length, 0.190-in. shank diameter, 0.255-in. thread o.d., 2.750-in. thread length, 0.625-in. head diameter SIP screw, 6-in. on center	0.131-in. x 2 1/2-in. nails, 6-in. on center	265	3:1
	0.131-in. x 2 1/2-in. nails, 4-in. on center 7/16-in. x 3-in. OSB Surface Spline	10-in. length, 0.190-in. shank diameter, 0.255-in. thread o.d., 2.750-in. thread length, 0.625-in. head diameter SIP screw, 4-in. on center	0.131-in. x 2 1/2-in. nails, 4-in. on center	330	3:1
	0.131-in. x 2 1/2-in. nails, 2-in. on center, two rows staggered 3/8-in. 7/16-in. x 3-in. OSB Surface Spline	10-in. length, 0.190-in. shank diameter, 0.255-in. thread o.d., 2.750-in. thread length, 0.625-in. head diameter SIP screw, 3-in. on center	0.131-in. x 2 1/2-in. nails, 2-in. on center, two rows staggered 3/8-in.	575	3:1

For **SI:** 1 inch = 25.4 mm, 1 PLF = 14.59 N/m

[1]Surface or block spline only at interior panel-to-panel joints. Specified fasteners are required on both sides of panel joint through the top surface only, as shown in Figure 3b.

[2]Interior support spline shall be solid lumber 1 1/2-inch-wide minimum and have a specific gravity of 0.42 or greater. Specified fasteners are required through both facings as shown in Figure 3c.

[3]Attachment of panels to interior supports is the responsibility of the designer and are not included with the shear strength capacities in this table.

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FIGURE 1—SIP SPLINE TYPES

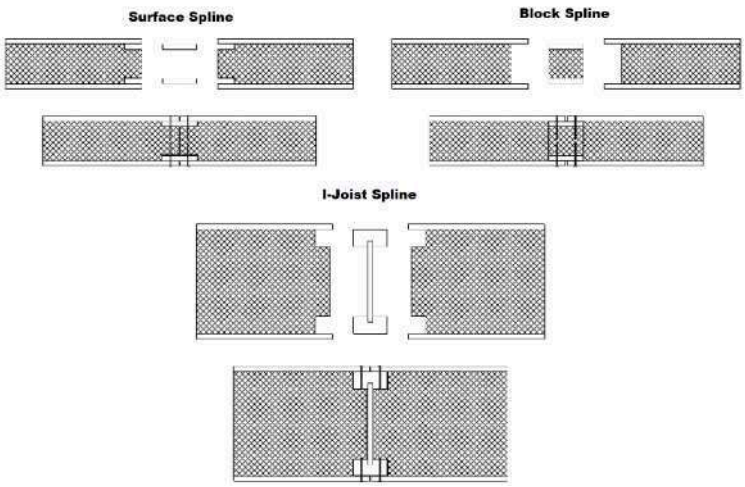

FIGURE 2—SIP SPLINE ATTACHMENTS

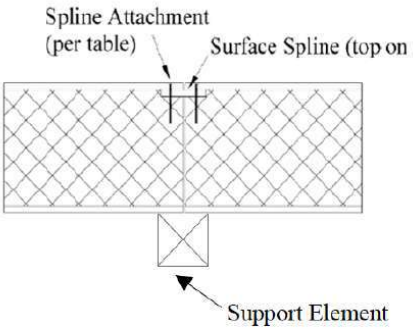

FIGURE 3A—INTERIOR SUPPORT SPLINE

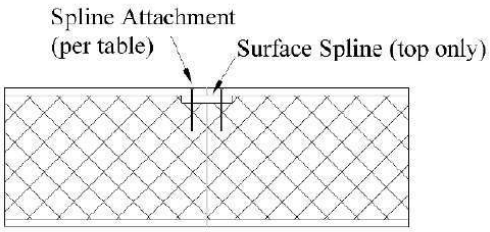

FIGURE 3B—SURFACE SPLINE

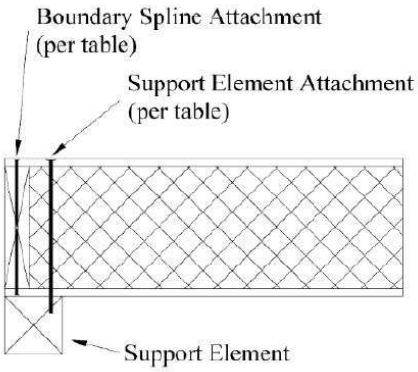

FIGURE 3C—BOUNDARY SUPPORT ELEMENT

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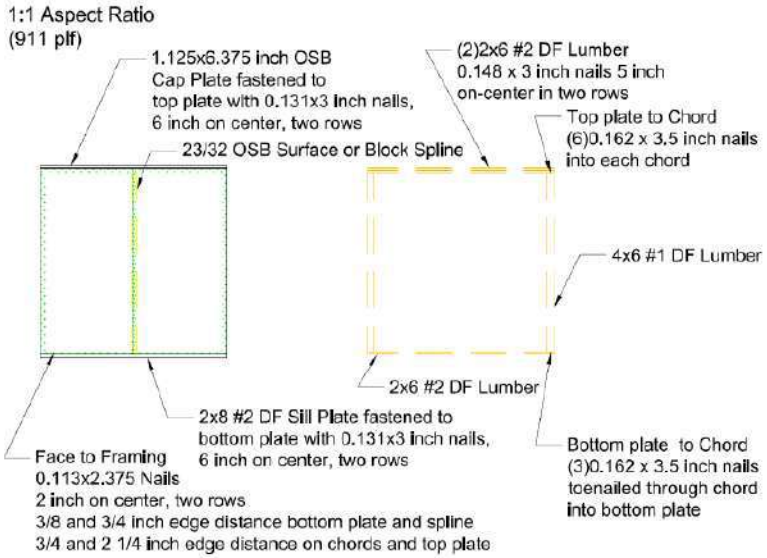

FIGURE 4 TABLE 13 SHEARWALL CONSTRUCTION (911 PLF)

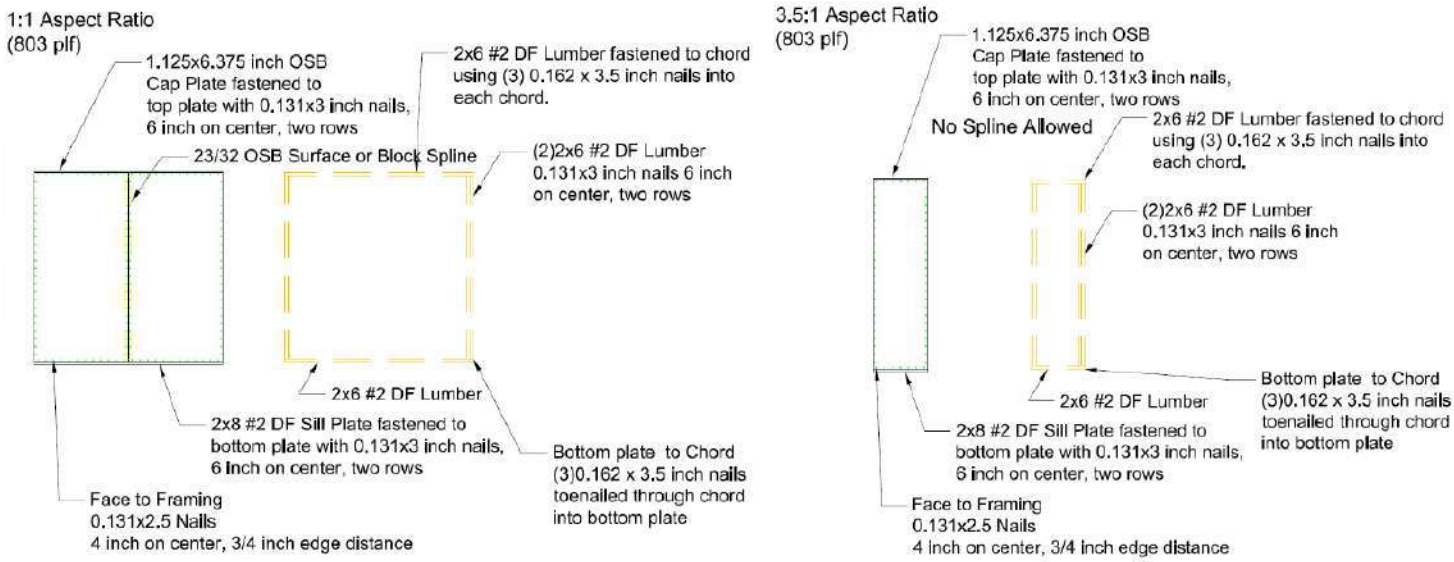

FIGURE 5 TABLE 13 SHEARWALL CONSTRUCTION (803 PLF)

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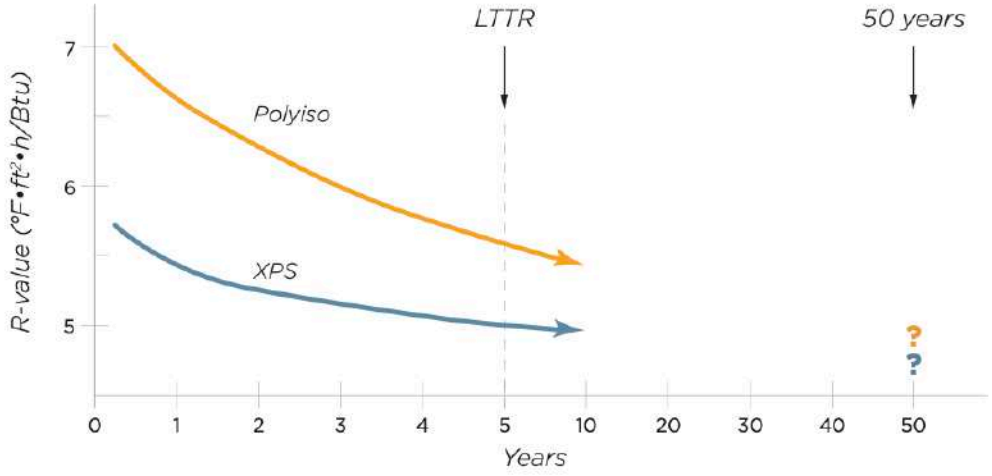
XPS & Polyiso Long-Term Thermal Resistance & R-Value Performance

Insulation is of paramount importance to reduce the energy consumption of buildings. In the summer, insulation reduces the heat flow from the hot exterior to the cool interior environment. In winter, insulation reduces the heat lost from the warm interior to the cold exterior. The resistance to heat flow through an insulation is called "thermal resistance". Thermal resistance is commonly referred to as R-value. A complete understanding of the R-value of insulation over its lifetime is critical to designing buildings that achieve reduced energy consumption. Most people understand that the higher the R-value, the greater the insulating power of an insulation. However, many people do not understand that the R-value of polyiso and XPS is lost over the lifetime of the product.

Long-term thermal resistance (LTTR) test methods have evolved considerably in the past decade. Two test methods are commonly used to report R-value for materials with trapped gases other than air. The test methods are ASTM C1303 and CAN/ULC-S770. Both test methods provide a method to estimate the long-term thermal resistance or long-term R-value of insulations. Each of the methods involves cutting thin sections approximately 3/8" (10 mm) from a sample of thicker insulation. Due to the thin size of the samples, diffusion of air in and trapped gases out is quicker than for the original thicker sample. The R-value loss of the thin sections can then be used to predict the R-value loss of the original thickness material.

Polyiso & XPS R-Values Over Time



50 YEAR R-VALUE

Most insulation users are interested in a true long-term thermal R-value for their insulations. A time period appropriate for building application is 50 years. In order to avoid confusion with the existing LTTR numbers commonly published, we recommend the use of a 50 year R-value be used for insulation specifications. A 50 year R-value is a more suitable long-term R-value for use in building design. The 50 year R-value can easily be determined using the existing protocol described in ASTM C1303 or CAN/ ULC-S770.

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Polyiso and XPS lose R-value over their lifetime. It is well understood that polyiso and XPS insulations trap gas in their cells other than air and polyiso and XPS will lose the gas over time. This is a natural process of materials coming to equilibrium with the environment. If this encapsulated gas assists with providing R-value then the R-value of the insulation will drop over time. The trapped gases in the cells of polyiso and XPS foam assist to provide an initial high R-value. During the life of these foams, air from the atmosphere diffuses in and the trapped gases diffuse out. The result is polyiso and XPS lose R-value over their lifetime. Unlike polyiso and XPS, the R-value of EPS is permanent.

The LTTR value commonly published from testing to ASTM C1303 or CAN/ULC-S770 is an estimate for the R-value of the insulation after five years. Many insulation manufacturers are promoting LTTR without providing a clear understanding that LTTR is an estimate for the R-value of the material after only five years. The concept of a five year R-value being equal to the "time-weighted 15 year average" is also often used by polyiso and XPS manufacturers. This approach assumes that the higher R-value established in years 1-4 is weighted by the inevitably lower R-value of the insulation in years 6-15. Neither the five year R-value, nor the time-weighted 15 year average approach is appropriate for use in building design. This is due to the fact that the R-values of polyiso and XPS continue to decline below the LTTR published five year numbers. Starting in year five and for the remaining life of the insulation, the R-values of polyiso and XPS are below LTTR published R-values.

EPS provides a permanent lifetime R-value. The five year and 50 year R-values for EPS are the same as the initial R-value since the gas trapped in the cells of EPS is atmospheric air. Unlike polyiso and XPS, EPS does not lose R-value over time. This is one reason the National Institute of Standards & Technology (NIST) chose EPS as the Standard Reference Material for Thermal Conductivity (SRM 1453), using a lot of samples certified in 1996 that still provide the same R-value today.

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